GEOTECHNICAL INVESTIGATION REPORT

PROPOSED (G+1 VILLA)

AT PLOT NO. 309010298, KHOR AL MUAIREEDH, RAK, U.A.E.

MR. AYMEN BIN HAMADI ESKANDARANI.

FOR

SAVANA ENGINEERING CONSULTANT

CST-RAK-2026-9922

JANUARY- 2026

Rev.	Date	Description	Author
00	18/01/2026	Geotechnical Investigation Report	Consult Laboratory
01	19/01/2026	Second option Recommendation	Consult Laboratory

- **ACCREDITED BY EMIRATES NATIONAL ACCREDITATION SYSTEM (ENAS)**
- **REGISTERED BY RAS AL KHAIMAH MUNICIPALITY**

ENAS
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Testing - NAL 225

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CONSULT SOIL TESTING LAB

MR. AYMEN BIN HAMADI ESKANDARANI.

18th JANUARY 2026

CST-RAK-2026-9922

SUBJECT: GEOTECHNICAL INVESTIGATION FOR PROPOSED (G+1 VILLA) AT PLOT NO. 309010298, KHOR AL MUAIREEDH, RAK, U.A.E

OWNER: MR. AYMEN BIN HAMADI ESKANDARANI

Dear Sirs,

CONSULT SOIL TESTING LABORATORY is pleased to submit this report of the geotechnical investigation for the **(G+1 VILLA), AT PLOT NO. 309010298, KHOR AL MUAIREEDH, RAK – UAE.**

This report presents the results of the field and laboratory test results, geotechnical analysis and interpretation of the findings, conclusions and recommendations to aid design and construction of the foundations.

We would like to take this opportunity to thank you for your confidence and look forward to be of service to you in the near future.

Sincerely yours,

CONSULT SOIL TESTING LABORATORY

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1. INTRODUCTION

The Geotechnical Investigation and Soil Testing phase of foundation engineering still involves some degree of uncertainties. No matter how extensive it is, there still is a doubt about its accuracy. Engineers attempt to compensate for these uncertainties by applying factors of safety in the analysis but unfortunately, this solution also increases the cost of construction due to over safe design.

In the effort of necessary level of conservatism in the Foundation design, the Geotechnical Engineer may choose more extensive Soil Investigation and Testing Program to better define the soil characteristics. The additional costs of such efforts will result in decreased construction costs. However, at some point, it becomes a matter of diminishing returns and eventually the increased cost of additional Soil Investigation and Testing does not produce corresponding reduction in construction costs. There is always optimum level of Soil Investigation and testing which gives the minimum cost of construction by providing the most economical Foundation Design.

Although there are times soil mechanic techniques can be applied to rock mechanics problems and vice versa but any such sharing must be done cautiously.

This Soil Investigation report was made according to following M/s. RAK Municipality Regulation.

2. THE OBJECTIVES OF SITE INVESTIGATIONS

The objectives of Soil exploration and characterization program include:

- Determining the location and thickness of soil and rock strata.
- Determining the location of the Ground Water Table.
- Recovering samples for testing and evaluation.
- Conducting tests, either in the field or in the laboratory to measure relevant engineering properties.
- Defining special problems and concerns.

3. PROJECT DESCRIPTION

The project involves the construction of **Proposed (G+1 VILLA MR. AYMEN BIN HAMADI ESKANDARANI., on plot no. 309010298, KHOR AL MUAIREED, RAK, UAE**. The work was carried out for Two (2) boreholes to depth of Eight (8) meter each upon the **Total Area of the plot is 450.00m² as per Ras Al Khaimah Structural Guidelines.** The site plan of the boreholes is shown in Drawing.No.1(Appendix-A).

4. ASSESSMENT OF FOUNDATION SOIL

The process of identifying the layers of deposits that underlie the proposed structure and their physical characteristics is generally referred to as subsurface exploration.

The purpose of sub-surface exploration is to obtain information that will aid the Geotechnical engineer to perform the following:

(a) Selecting the type and depth of foundation suitable for given structure.

(b) Evaluating the load bearing capacity of foundation.

(c) Estimating the probable settlement of a structure.

(d) Determining potential foundation problems due to the nature of the soil such as expansive soil, collapsible soil, and Sanitary landfill or dredge material.

(e) Determining the location of the Ground Water Table.

(f) Predicting lateral earth pressure for structures such as retaining walls, sheet pile bulkheads and braced cuts.

(g) Establishing construction methods for changing subsoil conditions.

Subsurface exploration may also be necessary when additions and alterations to existing structures are contemplated.

5. SCOPE OF WORK

The scope of works consists of the following:

- Making inspection visit to the site to collect information about the present land use, surface topography, geological features and surface drainage.
- Drilling of **2** boreholes down to a depth of **8.0m** each **as per Ras Al Khaimah Structural Guidelines**, and sampling of disturbed and undisturbed samples.
- Carrying out necessary field and laboratory tests.
- Performing engineering analysis of fields and laboratory findings.
- Developing conclusions and recommendations for foundation design and construction

6. METHOD OF INVESTIGATION

- **Field Investigation**

Fieldwork commenced on **JANUARY 14th, 2026** and was completed on **same day**. The scope of the work comprises the drilling of **two** boreholes **(BH1 & BH2)** to depth of **8.0m** each **as per Ras Al Khaimah Structural Guidelines**. The borehole locations are shown on the Site Plan in Appendix A.

A Drilling Rig Machine (**rotary**) was used for drilling the boreholes adopting **rotary** drilling method.

Using procedures specified in the code of practice for site investigation BS 5930:2015, Disturbed and split spoon samples were obtained from the boreholes for soil classification and laboratory testing.

- **Standard Penetration Test**

In order to determine the relative density of the revealed strata, Standard Penetration Test at frequent intervals of depth were conducted in accordance with BS 1377: Part 9:1990 "Methods of test for soils for civil engineering purposes."

The SPT consist of driving a 50mm external diameter thick walled tube (Split spoon sampler) into the bottom of the borehole using a 63.5 Kg hammer falling freely through 760mm.

Initially the sampler is driven 150mm into the soil to be seated and to pass through disturbed soil at the bottom of the borehole. The number of blows required for driving the sampler a further 300mm is recorded and termed as the "N" value. The results are shown on the attached borehole logs in Appendix B.

- **Laboratory Testing**

Soil and water samples were tested in accordance with BS 1377 (1990) "Methods of Test for Soils for Civil Engineering Purposes."

Laboratory testing consisted of a visual classification on all the soil samples. Particle size distribution and chemical analysis of soil were conducted on selected samples.

- **Particle Size Distribution**

Particle size distribution was carried out in accordance with B.S. 1377: 1990 Part 2: Method 9 "Determination of Particle Size Distribution".

Soil samples were mechanically analyzed by wet sieving for classification. The results are presented in the form of particle size distribution curves in Appendix C.

- **Chemical Analysis**

The likelihood of deterioration of the foundation concrete to aggressive in-situ condition was assessed by the determination of the pH, sulphate as Sulphur trioxide, and chloride content of the soil and ground water samples in accordance with the following B.S. Standards:

B.S.1377: 2018: Part 3: Clause 7.3/7.6 "Determination of the Sulphate Content of soil and Groundwater."
B.S.1377: 2018: Part 3: Clause 9.2 "Determination of the Chloride Content."
B.S.1377: 2018: Part 3: Clause 12 "Determination of the pH Value.

Chemical Tests are presented in Appendix-

7. REGIONAL GEOLOGY AND WEATHER CONDITIONS

The geology of the United Arab Emirate, and Arabian Gulf Area, has been substantially influenced by the deposition of marine sediments associated with numerous sea level changes during relatively recent geological time, with the exemption of mountainous regions shared with Oman in the North-East, the country relatively low-lying with near surface geology dominated by Quaternary to late Pleistocene age, mobile Aeolian dune Sands and Sabhkha/ evaporates deposits.

The site is situated in **Ras Al Khaimah** where a hot arid climate prevails. A hot arid climate is one where evaporation exceeds precipitation such as rain, snow and dewfall. This climate regime produces characteristics hot dessert terrains. Average annual rainfall may only be a few centimeters (even only a few millimeters in some parts) which usually occurs seasonally and sometimes only for single cloudburst. Summer shade temperatures are frequently in excess 40°c and humidity maybe very high near the coast. The contrast between maximum night and day temperatures and between night and day humidity is often great. Strong persistent winds are normal in many areas. This unfavorable climate imposes adverse on the concrete structures such as:

- High temperatures and seasonal changes
- High humidity and change in relative humidity
- Strong shifting winds during day time
- Condensation at night due to low temperature
- Windborne salt laden dust storm
- High solar radiation day time

8. FIELD WORK

- **Drilling**

Two boreholes were drilled on **14th of JANUARY 2026,** down to **a depth of 8.0m** each below the existing ground surface upon the **Total Area of the plot is 450.00m² as per Ras Al Khaimah Structural Guidelines**.
The drillings were executed by rotary Drilling Rig using rotary Drilling Method. The Borehole Logs are presented in Appendix B.

Plot Number	Borehole Number	Borehole Surface Level (ER.LVL)	Drilled Depth	UTM-Coordinates		Field Work Date (dd/mm/yy)
				Northing	Easting	
309010298	BH-1	(- 0.50)	8.00 m	25.825205,55.970771		14/01/2026
	BH-2	(- 0.50)	8.00 m	25.825307,55.970870		14/01/2026

- **Sampling**

Soil Disturbed, undisturbed and Split Spoon samples were obtained from the boreholes. The soil samples were placed in airtight plastic bags, and then transferred to the laboratory for further testing.

9. SUBSURFACE CONDITION AND CAVITY CONDITION

The subsurface conditions encountered at the borehole locations have been summarized in the borehole logs in Appendix B.

Based on the drilled boreholes, no open cavities were identified and non-free fall of drilling tool was observed during the drilling. This observation was covered only in the location of boreholes and at drilled depth.

10. FIELD TESTING

Standard Penetration Test (SPT)

- It was developed in the late 1920's and has been used extensively through out the world because of this long record of experience; the SPT is well established in engineering practice. The test procedure was standardized only in 1958 when ASTM Standard D 1586 first appeared.

- Although SPT is plagued by many problems that affect its accuracy and reproducibility, it is continued to be used, primarily because of its low cost and increased familiarity with it. Even after standardization, the test has a poor repeatability.

- Standard penetration Tests (SPT) was performed at various depths in the boreholes to asses the relative densities of the ground materials. The tests were performed in accordance with BS 1377: 1990 Part 9, "Determination of Penetration Resistance using Split Barrel Sampler (SPT) or ASTM: D 1586.

- The SPT consists of driving a Standard 50mm outside diameter thin wall sampler into soil at the bottom of a borehole, using repeated blows of a 63.5kg hammer falling 760mm. The SPT N value is the number of blows required to achieve a penetration of 300mm, after an initial seating drive of 150mm.

- The test results are shown on the boring logs at the respective test depths. Interpretation of the SPT test results can be found in the Legend of Boring Logs (Appendix B)

11. GROUND WATER

Observation concerning ground water were made during and at completion of the drilling operations.At the time of investigation, the ground water level was established at a depth of **3.40m** below working level (BH.1 to BH.2)**.** However, this level may be subject to seasonal & tidal variations or changes if **de-watering** takes place in the vicinity . Reconfirmation of the ground water regime is recommended before the start of any work

Plot Number	Borehole Number	Borehole Surface Level (ER.LVL)	UTM-Coordinates		Groundwater Depth (m) (BEGL)	Groundwater Level (ER.LVL)
			Northing	Easting		
309010298	BH-1	**(- 0.50)**	25.825205,55.970771		**-3.40**	**-3.90**
	BH-2	**(- 0.50)**	25.825307,55.970870		**-3.40**	**-3.90**

12. REVIEW OF LITERATURE AND THEORIES FOR THE DESIGN OF FOUNDATIONS:

Geotechnical Parameters for Design of Shallow Foundations:

Proper selection of foundation members, dictates their being capable of sustaining the structural loads and transmitting these loads safely to the supporting ground, so it must provide for two points. One is to avoid foundation soil failure, which leads to structural collapse, and the second is to prevent excessive settlement, which may lead to restricting the possibility of using the structure.

Terzagi's equation is one the most widely used equations to calculate bearing capacity for Shallow Foundations. Despite that it was originally developed for soils, it is also used to calculate the bearing capacity for foundations on rocks provided properly selected factors are used. This equation is of the form.

$$q_{ult} = C\ Nc\ Sc + q`Nq + 0.5\ BN\gamma S\gamma$$

Where:

q_{ult}	=	Ultimate Bearing Capacity
$Sc, S\gamma$	=	are shape factors
q	=	γh
C	=	Cohesion of Soil
N_c, N_q, N_γ	=	are factors related to angle of internal friction (Φ)

On the other hand the settlement has to be within certain limits, and this may dictate some limitations on the permissible bearing capacity which is obtained through applying a factor of safety 3 on the ultimate bearing capacity.

- For sandy soils, the settlement occurs as the load is applied and there are no time dependent effects. Under these conditions, settlement can be calculated using Elastic Theory by using appropriate values for the Young's Modulus and Poisson's ratio of the soil mass.

13. GENERAL DISCUSSION FOR THE CHOICE OF SUITABLE FOUNDATIONS

In designing foundations, the engineer must satisfy two independent foundation stability requirements, which must be met simultaneously:

1. There should be an adequate safety against shear failure within the soil mass. In other words, the working loads should not exceed the allowable bearing capacity of the soil being built upon.

2. The probable maximum and differential settlements of the soil under any part of the foundations must be limited to safe and tolerable limits.

The choice of particular type of foundation depends upon the character of the soil, the presence of ground water at the site, the magnitude of the imposed loads, and the project characteristics. One has to choose the type of foundation which is not merely safe but also economical.

For the particular case, the following prevailing load and site conditions exist:

1. The imposed loads from the proposed structures on the foundation ground are expected to be light to medium due to the nature of the proposed structures.
2. Ground water was encountered (-**3.40m**) down to the drilled depth.

3. The materials encountered along with field and laboratory test results are shown in Appendix C and logs of borings in Appendix B.

According to the above conditions, shallow foundation **(isolated footings or strip footings or Raft)** can be used to support the proposed structures as per the following recommendations.

14. CONCLUSIONS AND RECOMMENDATIONS BASED ON INVESTIGATION RESULTS

Based upon the Borehole Logs, Field and routine Laboratory tests results and position of the Ground Water Table and nature of the project, is It is recommended to perform one of the following methods in order to have a method can be applied for this case, taking in the consideration the economically and feasibility factors.

NOTE: since the site is closer to the water front area, the ground water encountered may vary due to the tidal variations. De watering shall be done, if water is found above excavation level during the time of construction.

Option 01:

- In order to lay the shallow foundation, proceed as follows:

At the time of investigation, level of the Site was **approximately (2.65)** with respect to the Municipality Datum.
Design Level (D.LVL) of the site is at **3.40** as per the Level Difference certificate.
[Level Difference Certificate (Elevation) is attached in Appendix A.]

- Excavate and level the existing soil at the Level of **-2.50m** below the **D.LVL (3.40)** level established on the site.
- Reinforce the base of excavation by using crushed rocks or boulders of size 75mm to 150mm. The rock filling shall be compacted by heavy vibratory roller until no further penetration of rock fill in to the soil during compaction is achieved. The rock fills to be extended to **0.50m** above excavation level.
- Place **Two (2)** layers of selected granular backfill materials **(Road Base) 25cm** thick to reach the required foundation level. Each layer shall be compacted by heavy vibratory roller to a degree of compaction not less than **98%** of the maximum dry density as obtained by modified proctor compaction test (BS-1377 PART-2:2022).
- Plate bearing test shall be carried out at foundation level as quality control measure to verify the required allowable bearing pressure and total settlement criteria under foundations.
- **A 100 mm** thick layer of blinding concrete (PCC) should be placed to protect the final compacted surface.
- Shallow foundation can be placed on engineered fill. (i.e., Approximately **-1.40m** below the **D.LVL (3.40)** established on the site.)
- Foundation level will be at **2.00** with respect to the Municipality Datum.

TABLE 14.1
NET ALLOWABLE BEARING CAPACITY FOR ISOLATED/ STRIP/COMBINED FOOTINGS

Width of Footing	Net Allowable Bearing Pressure	Modulus of sub-grade
Up to 2.00m	120 kN/m2	14,400 kN/m3
Up to 3.00m	110 kN/m2	13,200 kN/m3
Up to 4.00m	100 kN/m2	12,000 kN/m3
Up to 5.00m & Wider	90 kN/m2	10,800 kN/m3

Raft Foundation	**140 kN/m2**	**8,400 kN/m3**

The foundation settlement related to these values of pressure is less than the tolerable limits of **1 inch (25mm)** for the isolated and **2 inch (50mm)** for Raft foundation.

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Option 02:- In order to lay the Raft foundation, proceed as follows:

- Excavate and level the existing soil at the Level of **-2.25m** below the **D.LVL (3.40)** level established on the site.
- Reinforce the base of excavation by using crushed rocks or boulders of size 75mm to 150mm. The rock filling shall be compacted by heavy vibratory roller until no further penetration of rock fill in to the soil during compaction is achieved. The rock fills to be extended to **0.50m** above excavation level.
- Place **One (1)** layer of selected granular backfill materials **(Road Base) 25cm** thick to reach the required foundation level. The layer shall be compacted by heavy vibratory roller to a degree of compaction not less than **98%** of the maximum dry density as obtained by modified proctor compaction test (BS-1377 PART-2:2022).
- Plate bearing test shall be carried out at foundation level as quality control measure to verify the required allowable bearing pressure and total settlement criteria under foundations.
- **A 100 mm** thick layer of blinding concrete (PCC) should be placed to protect the final compacted surface.
- Shallow foundation can be placed on engineered fill. (i.e., Approximately **-1.40m** below the **D.LVL (3.40)** established on the site.)
- Foundation level will be at **2.00** with respect to the Municipality Datum.

TABLE 14.2
NET ALLOWABLE BEARING CAPACITY FOR RAFT FOOTINGS

Raft Foundation	**130 kN/m2**	**7,800 kN/m3**

The foundation settlement related to these values of pressure is less than the tolerable limits of **1 inch (25mm)** for the isolated and **2 inch (50mm)** for Raft foundation.

- Adequate cover of backfill **(minimum 0.5m**) shall be provided above the top of the foundation to protect the foundations ground from erosion and seasonal weather variation

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Option 03: Ground Improvement by Vibro compaction

Vibro compaction is a ground improvement technique that densifies clean, cohesionless granular soils by means of a downhole vibrator. The vibrator is typically suspended from a crane and lowered vertically into the soil under its own weight. Penetration is usually aided by water jets integrated into the vibrator assembly. After reaching the bottom of the treatment zone, the soils are densified in lifts as the probe is extracted. During vibro compaction, clean sand backfill is typically added at the ground surface to compensate for the reduction in soil volume resulting from the densification process. The vibratory energy reduces the inter granular forces between the soil particles, allowing them to move into a denser configuration, typically achieving a relative density of 70 to 85 percent. The vibro compaction penetration spacing is between 6 feet and 14 feet, with centers arranged on a triangular or square pattern. Compaction takes place without setting up internal stresses in the soil, thus ensuring permanent densification.

Option 04: Ground Improvement by Vibro Replacement (Stone Columns)

Since the subsoil strata are loose layers the stone columns techniques are suitable for ground Improvement work. The vibro replacement method is used to produce a stone column, below the ground level by sinking a vibratory probe that initially creates a punch down to the bottom of the stratum to be improved.The punch is then filed in stages with granular materials (commonly ranging in gradation from 2mm to 50mm). Each lift the vibro flot is lifted up and granular material introduced at the top and allowed to slip down around the probe and the lift compacted as the probe is lifted and lowered successively and until the column of high compactness and average dimensions is formed. Achieving the designated columns at the designated grid spacing ensures that the overall level of improvement and hence the required load carrying capacity is achieved and as confirmed by standard in situ testing.

The depth of stone column shall be exceeded that of the very loose and soft layers below the existing ground surface. The distribution of these columns below the foundation ground (i.e. spacing, diameter and depth) shall be decided by specialist and experienced contractors (normally 1.2m to 3.0m spacing and 0.5m to 0.75m diameter, and typically 1.0 to 2.0m m into the dense/stiff soil) to achieve the required ground bearing pressures for raft/Isolated foundation.

* Stone column techniques suggest that bearing capacities of (160 kN/m²) or more can be achieved depending on the design adopted for ground improvement and type of soil existing in situ.

Enough care should be taken do not cause excessive settlement to the neighboring structures due to the vibration probes. Plate load tests shall be carried out to assess the improvements attained and to confirm the required bearing pressures values that can be allowed after the improvements by this technique.

Vibro-compaction is a ground improvement technique that densifies clean, cohesionless granular soils by means of a downhole vibrator. The vibrator is typically suspended from a crane and lowered vertically into the soil under its own weight. Penetration is usually aided by water jets integrated into the vibrator assembly. After reaching the bottom of the treatment zone, the soils are densified in lifts as the probe is extracted. During vibro-compaction, clean sand backfill is typically added at the ground surface to compensate for the reduction in soil volume resulting from the densification process. The vibratory energy reduces the inter granular forces between the soil particles, allowing them to move into a denser configuration, typically achieving a relative density of 70 to 85 percent. The vibro-compaction penetration spacing is between 6 feet and 14 feet, with centers arranged on a triangular or square pattern. Compaction takes place without setting up internal stresses in the soil, thus ensuring permanent densification.

Option 2-Ground Improvement by MicroPile Technique (None working Pile)

Shallow foundations (isolated footings and/or strip footings) can be used to support the structural loads on the improved area with Micro Piling. MicroPile is a small-diameter, drilled and grouted non-displacement micropile that is typically reinforced. a micropile is constructed by drilling a borehole, placing steel reinforcement, and grouting the hole. The distribution of these MicroPiles below the foundation ground (i.e. diameter and depth) shall be decided by specialist and experience contractors to achieve the required ground bearing pressures. Foundation level can be -1.15 m below BM Level.

Recommendation: Shallow Foundations on Improved Ground Using Micropiling Technique (As per BS EN 14199)

Where soil conditions are inadequate to support shallow foundations directly (e.g., soft clay, loose sand, fill), it is recommended to improve the ground using micropiles to enhance bearing capacity and reduce settlement. Micropiles act as reinforcement to the subsoil, enabling the safe use of shallow footings such as rafts or pad foundations.

Micropiles may be used for:-

working under restricted access and/or headroom conditions;

Foundations of new structures (particularly in very heterogeneous soil or rock formations);

Reinforcing or strengthening of existing structures to increase the capacity to transfer load and acceptable load settlement characteristics, e.g., underpinning works;

Reducing settlements and/or displacements;

Design and Execution Standards

➤ **BS EN 14199:2015 (Micropiles – Execution of Special Geotechnical Works)**

Micropiles should be designed for axial load transfer through skin friction and/or end bearing.

The micropile layout (grid spacing, depth, diameter) must be based on geotechnical investigation results.

Installation methods (e.g., drilled) and grouting technique must be selected based on ground conditions and performance needs.

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SOIL PARAMETERS

The following soil parameters (coefficient at rest, active and passive pressures, dry density and the angle of friction resistance) are estimated at depth as given below based on the standard penetration test (SPT) values and laboratory test. The coefficient of earth pressure is calculated by Rankine's Method. The recommended soil parameters are for different locations within the first 3m depth. The following soil parameters may be adopted for the design of shoring system:

Soil Parameters	**Depth Range**
	R.L. -0.00m to R.L. -3.00m
Angle of Shearing Resistance	31°
Unit Weight of Soil (above water table) (kN/m^3)	17
Earth Pressure Coefficients	
• Active earth pressure coefficient (k_a)	0.33
• Passive earth pressure coefficient (k_p)	3.07
• Earth pressure at rest (k_0)	0.49

Soil Parameters	**Depth Range**
	R.L. -4.00m to R.L. -8.00m
Angle of Shearing Resistance	31°
Unit Weight of Soil (above water table) (kN/m^3)	17
Earth Pressure Coefficients	
• Active earth pressure coefficient (k_a)	0.32
• Passive earth pressure coefficient (k_p)	3.11
• Earth pressure at rest (k_0)	0.49

15. STRUCTURAL AND GENERAL FILL MATERIALS SPECIFICATAIONS

The materials which will be excavated from the site consist of fine to medium grained silty shelly SAND. Therefore, the materials based on classification test (sieve analysis) are considered suitable for backfilling purposes. However, the final decision shall be made during construction.

❖ Type A Fill Material

Type A Soil replacement and backfilling using imported structural fill material under the light weight modules should comply in general with the following specifications, as per the soil report recommendations.

- Imported structural fill material to be used in such locations shall be classified as A-1-b according to ASTM D 3282 classification.
- Well grade material with maximum size of 75mm.
- The Fraction of fines passing sieve No. 200(0.0075-mm) shall be less than 15%.
- Free organic materials.
- Plasticity Index (P.I.) of material shall be less than 6%.
- Maximum Dry Density shall be greater than 1.7g/cm3.
- Minimum CBR value of 50%.
- In addition to the specifications mention in the soil report, imported structural fill shall be free of any organic material and shall show minimal percentages of chloride and sulfate when tested for chloride and sulfate contents.
- Fill should be place in loose layers not exceeding 250mm thickness, and each layer should be compacted with suitable equipment to a minimum 95% of its Proctor maximum dry density.
- It is recommended that confirmation of compliance with compaction requirements be made by field density testing at a reasonable frequency.
- Usual type of significant constituent material should be stone fragments, gravel and sand.

❖ Type B Fill Material

Type B fill material shall be used as fill only for backfilling to raise the site level to approve finish graded level, at locations where installation of structure or equipment is not planned and in open areas where only grading is required. As per the soil report recommendations.

- Fill material to be used in such location shall be classified as A-2-4 or A-2-5 according
- to ASTM D 3282 classification.
- In addition to the specifications mention in the soil report, imported structural fill shall
- be free of any organic material and shall show minimal percentages of chloride and sulfate when tested for chloride and sulfate contents.
- Imported structural material shall be compacted to the levels of compaction and
- thickness of compaction layers mentioned in the soil report, which vary according to the location of application in the plot, and shall also be tested as per recommendation of the soil report.

❖ Road Base Material

General Requirements

- Coarse aggregate retained on the 1.75-mm (No. 4) sieve shall consist of durable particles of crushed stone, gravel, or slag capable of withstanding the effects of handling, spreading and compacting without degradation productive of deleterious fines of the particles which are retained on a 9.5-mm (3/8-inch.) sieve, at 75 % shall have two or more fractured faces.
- Fine aggregates passing the 4.75-mm (No. 4) sieve shall normally consist of fines from the operation of crushing the coarse aggregate. Where available and suitable, additional of natural sand or finer mineral matter, or both, is not prohibited.
- The fraction of the final mixture that passes the 75-mm (No.200) sieve shall not exceed 60% of the fraction passing the 600-mm (No. 30) sieve. The fraction passing the 425-mm (no.40) sieve shall have a liquid limit no greater than 25 and shall have a plasticity index no greater than 4. The sand equivalent value of the fines aggregates shall be no lower than 35.
- For material to be used as a sub-base at a greater depth than probable frost penetration, the plasticity index and sand equivalent requirements are modified to a maximum of 6 and a minimum of 30 respectively.
- The gradation of the final composite mixture shall conform to an approved project specification.

❖ Compaction

Prior to the laying of the foundation, the exposed sub-grade material at the formation level should be dry leveled, clear from big stones and should be proof rolled by proper vibrator. After excavation the ground should be proof rolled by a suitable compactor to achieve a dry density of not less than 95% as the maximum dry density obtained by laboratory compaction. The backfill material shall be carefully laid and compacted. The fill material should be placed in layers of not exceeding 25.0cm. These layers should also be compacted to attain a dry density of not less than 95% as the maximum dry density obtained by modified proctor test (BS-1377-Part-2:2022 & ASTM D-1557).

16. EXCAVATION SIDE SLOPES AND / OR LATERAL SUPPORT SYSTEM:

Unsupported temporary excavation walls and slopes should be conducted at gradients that secure their stability. Unsupported temporary excavations in the different strata should consider the following gradients:

- In the very loose soils, slopes should have maximum gradients of *1V:2H.*
- In the medium dense to dense soils, slopes should have gradients of *1V:1.5H.*
- In weak bed rock, slopes should have maximum gradients of 1V:1H.
- Slopes for more than 3.0m deep excavation shall be designed by an engineer on a case by case basis.

In case the above mentioned gradients are not feasible due to the site specific restrictions steeper slopes can be adopted considering appropriate slope stabilization measures.

Vertical unsupported excavation walls in the bedrock can be achieved; however the maximum stable height should be determined for each particular case, taking into considerations the thickness of the topping soil, expected surcharge loads, geo-mechanical properties of the rock mass among other design parameters.

17. SIESMIC DESIGN PARAMETERS:

Shear wave velocity has been estimated on the co-relation related to SPT Values for the sand strata based on published literature for rock strata. Based on the average shear velocity and the classification based on the uniform building code 1997, the following seismic parameters shall be adopted.

- UAE has been grouped under **Zone (2A)** with seismic zone factor (Z) of 0.15.
- Soil profile type **S_E** representing **Soft soil profile** (with SPT "N" values <15) may be adopted for the design with a **shear wave velocity** of **180 m/sec.**
- A site factor (S- factor) of 1.00 must be used in the design.
- Generally, based on the SPT (N-Values): The soil profile types can be considered as tabulated below.

Soil/Rock Description	Shear Wave Velocity (m/s)	SPT Range	UCT Range (Kpa)	Soil Profile Type
Hard Rock	1500	-	-	SA
Rock	760 to 1500	-	-	SB
Very Dense Soil & Soft Rock	360 to 760	>50	100	SC
Stiff Soil Profile	180 to 360	15 to 50	50 to 100	SD
Soft Soil Profile	180	<15	50	SE
Soil Requiring Site -Specific Evaluation				SF

- **The seismic coefficient C_a = 0.22**
- **The seismic coefficient C_v = 0.32**

18. CONCRETE FOR FOUNDATIONS:

Based on the results of sulphate content determination test for the soils samples recovered from SPT which are given in Appendix-D, the recommendations for the cement type as per different references were presented in the report, the site is classified by **BS-8500-1-2015+A2-2019, BRE Special Digest 1:2005 "Concrete in Aggressive Ground" and CIRIA Report C577, BS EN 197-1**.
Any of the recommended cement type mentioned in Table **18.1** can be used for the underground structures and are suitable to the existing ground conditions. The recommendation on Table **18.1** refers to the minimum cement content and maximum cement ratios that can be used in the project were for guidance purposes only. The final decision should be taken by a qualified Engineer during construction and it should meet's the required project specification criteria whichever stricter.

The primary cause of serious deterioration in reinforced concrete is corrosion of the reinforcement, due to present attack by chlorides in concrete, either within concrete aggregates and mixing water, or by penetration from surrounding environment. Since chloride induced reinforcement corrosion can only occur in the presence of oxygen and water, the risk of corrosion can be reduced by control of chloride in concreting materials and by ensuring adequacy, integrity and impermeability of the concrete cover.

Sulphates attacks on concrete are caused by the presence of high sulphates contents either by the ingress of the sulphates present in the surrounding environment such as foundation soils and the groundwater, or by the presence of sulphates in the concrete ingredients themselves. The attack results in a considerable internal expansion which may lead to cracking and disintegration of the concrete. This effect can be reduced by the use of selected cements or by suitable protection of the concrete.

The results of chemical analysis presented on the summary sheet show the sulphate contents of the soil samples tested was 2058Mg/l. The chloride content for the soil tested was 900Mg/l. The pH value of the sample is 8.13.

Where sulphates and chlorides occur together in high concentrations, **sulphate resisting cement provides less protection against the reinforcement corrosion**. In such cases the test exposure conditions should be studied in conjunction with recommendations for concrete mix design.

Appendix F presents brief discussion on the chemical attack on buried concrete along with extracts of references for assessment of exposure conditions and concrete specification.

Classification of the severity of chemical attack in the investigated site was based on the foundation soil Sulphate content as well as the type of exposure conditions. Accordingly, the site has been classified as follows:

Table 18.1: Site Classification

Reference Document	Soil Class / Design Class or Exposure Condition	Cement Type/ Combination Group	Cement Content (kg/m3)	Maximum W/C
BS 8500-1 (Soil Class)	**DS-3**	**CEM I (OPC) or SRPC**	**320**	**0.45**
		CEM II/A, CEM II/B, CEM III/A, CEM III/B, CEM V	**380**	**0.45**
		CEM I + PFA (Fly Ash) (25-40%)	**380**	**0.45**
		CEM I + GGBS (Ground Granulated Blast-furnace Slag) (66-80%)	**340**	**0.45**
		CEM I + Silica Fume (5-10%)	**360**	**.45**
BRE Special Digest 1:2005	**DS-3** **AC-3s**	**F**	**340**	**0.50**
		E	**360**	**0.45**
		D, G	**380**	**0.40**
CIRIA Report C577	**b**	**-----**	**320**	**0.50**

*** When concrete is cast directly in contact with soil the minimum cover should be increased to 75mm.**
****This recommendation provided is not part of ENAS Accreditation.**

Modified or confirmed design recommendations shall be a responsibility of the designer, who shall finalize the concrete specification.

The concrete mix design and construction details shall be in accordance to the project specifications. The project specifications shall take precedence over the recommendations of this report.

Table 18.2: Foundation Concrete Recommendation

Location of foundation	Chloride Content in Concrete Surrounding	Design Chemical Class (DC Class)	
		Intended Working Life	
		At Least 50 Years	At Least 100 Years
Above Ground Water	Significant	DC-3(FND3)	DC-3(FND3)

Note: There is no widely accepted view of the concentration at which the chlorides become significant in soil or ground water, but limited experience in Gulf region suggests it may be as low as 500 mg/L, particularly in situation where alternative wetting and drying or capillary rise affect the concrete. (Ref. CIRIA Special Publication).

Additional Recommendations (as per BS 8500, Clause 6.3 & Annex A)

To enhance durability and long-term performance in aggressive sulfate + chloride environments:

- **Use Suitable Cement Combinations**
- **Prefer SRPC, or CEM I + GGBS (≥ 40%) or CEM I + PFA (≥ 25%)**
- **Must be tested and approved for DC-4 performance.**
- **Low Permeability Concrete**
- **Use superplasticizers to reduce W/C without affecting workability.**
- **Consider waterproofing admixtures.**
- **Increased Concrete Cover**
- **For XD3 exposure, ≥ 50–75 mm cover to reinforcement is required.**
- **Protective Measures**
- **Bituminous coating or membranes on foundations and buried concrete.**
- **Apply protective coatings to exposed concrete in contact with aggressive soil/water.**
- **Corrosion Mitigation**
- **Consider corrosion inhibitors or galvanized/reinforced bars in highly sensitive areas.**
- **Ensure chloride class Cl 0.40 is not exceeded by total chloride in mix.**
- **Proper Site Practices**
- **Good compaction, formwork sealing, and 7-day minimum curing.**
- **Prevent contact with stagnant or flowing groundwater during curing phase**

19. GENERAL COMMENTS:

In the absence of availability of full loading conditions imposed by the structure to be supported on the foundations it is not possible to decide the most appropriate analytic model for evaluating the interaction between the structural loads with their configuration and properties of the supporting soils and rock and such as the computations of parameters like total settlements, differential settlements and angular are not feasible.

Conclusions and recommendations made in this report are based on the findings from the drilled boreholes, and Laboratory tests results. Due to the limited extent of the soil investigation, it is most probable that some variation may be found at the time of execution of the project in the Sub – Strata encountered.

Most Engineers work with manufactured products that have very consistent and predictable engineering properties, but Geotechnical engineers do not have this facility. They work with soil and rock, which are natural materials whose engineering properties vary dramatically from place to place, for example, one site may be underlain by strong, hard deposits while another may be underlain by soft, weak deposits, and thus, instead of specifying required properties, Geotechnical engineer's task becomes to determine the properties of the existing soils.

The best way to deal with such uncertainties is continued monitoring of sub- surface during construction. Often new information becomes available during construction and if the new conditions are found to be different from the anticipated conditions, then the design may need to be changed accordingly even at the execution stage. In well managed projects, site characterization continues through out construction, period since further data often becomes available and may dictate changes in the design. Therefore, Geotechnical monitoring during construction is most essential and is highly recommended.

Design of Geotechnical structures involves a certain amount of uncertainty in the value of the input parameters which include the structural geology, material strengths, ground water pressures, floods and seismic events, reliability of the analytical procedure and construction methods. In view of these uncertainties and heterogeneous nature of the soils and rocks along with the creep phenomenon the recommendations and procedures contained in this report are intended to be used with caution, therefore, prior to their use in connection with any design, report or specifications they should be reviewed with regard to the full circumstances of such use.

APPENDIX A

Site Plan Showing Borehole Locations

بلدية رأس الخيمة
إدارة التخطيط والمساحة
Ras Al Khaimah Municipality
Town Planning and Survey Administration

المخطط الموقعي / ملك

بلديـــة رأس الخيمـــة
Ras Al Khaimah Municipality

المالك : ايمن بن حمادى اسكندرانى

مواصفات القسيمة

الرقم المميز : 309010298
اسم المنطقة : خور المعيريض
استخدام الارض : سكني

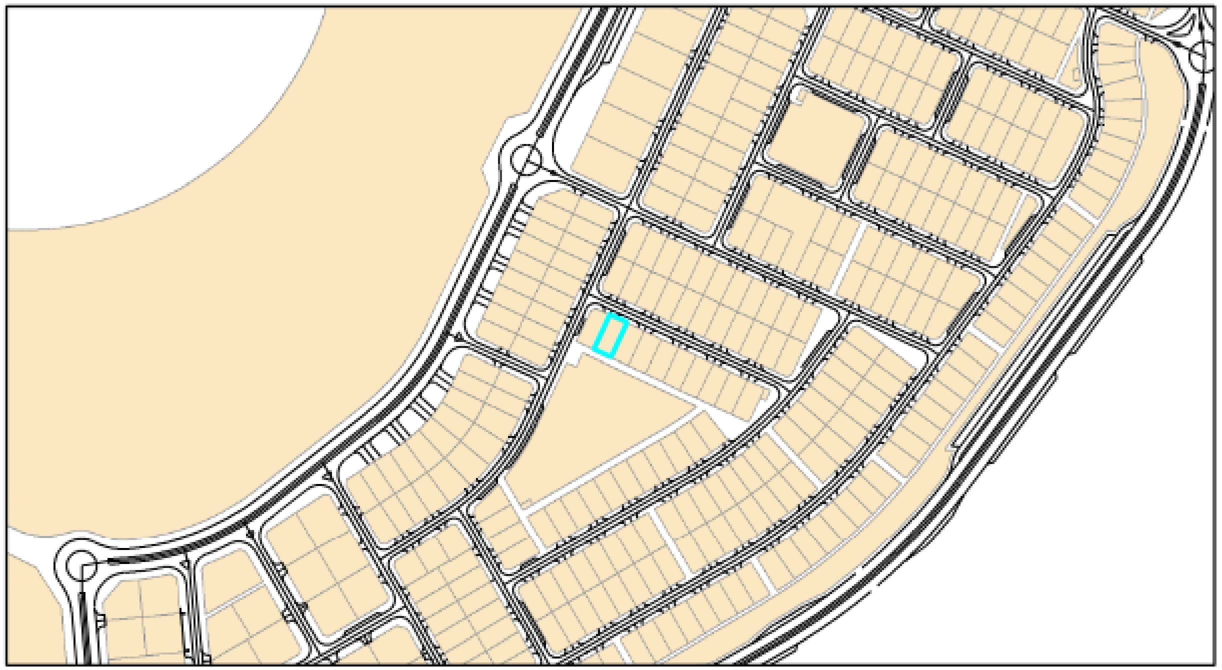
المساحة الكلية	450 متر مربع	=	4843.76	قدم مربع
المساحة المتأثرة	متر مربع	=		قدم مربع
المساحة المتبقية	450 متر مربع	=	4843.76	قدم مربع

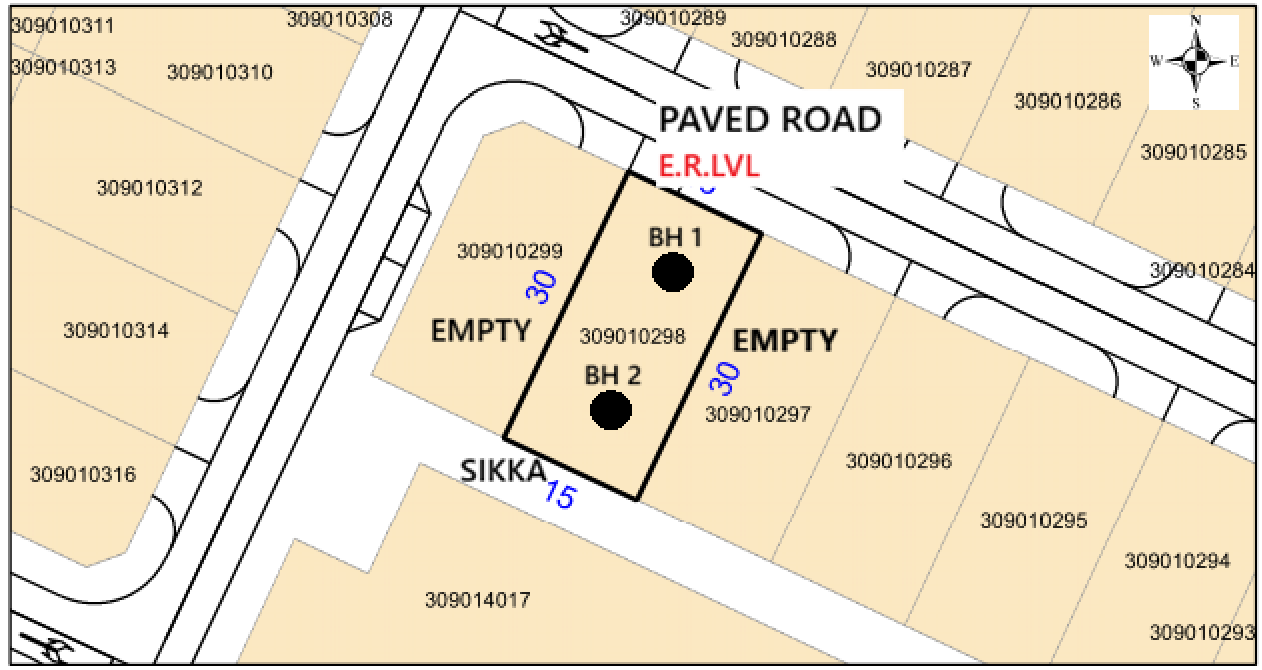
رقم المخطط : 21289
تاريخ المخطط : 02/01/2026
بدلا من المخطط رقم : 2014/38929
رقم الوثيقة : 2026/00003

الاشتراطات التخطيطية

نسبة البناء	عدد الطوابق		منسوب الرصيف
	الارتدادات		
الشمال	الجنوب	الشرق	الغرب
	البروزات		
الشمال	الجنوب	الشرق	الغرب

ملاحظات: أرض خالية





عند البيع او الشراء يرجى مراجعة ادارة التخطيط والمساحة بشأن التعديلات التي طرأت على المخطط

اعتماد مدير ادارة التخطيط والمساحة

Government of Ras Al Khaimah
حكومة رأس الخيمة

Ras Al Khaimah Municipality
بلديـــــة رأس الخيمـــــــــة

شهادة منسوب تصميمي

التـاريخ: 20/01/2026
الاشارة : 3144162

المنسوب التصميمي : 3.4

WGS 84

CoordinateNSF

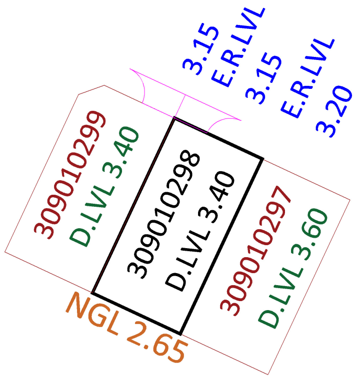
الاسم: ايمن بن حمادى اسكندرانى

بطاقة الهوية : 3000176730

رقم القسيمة : 309010298

المنطقة: KHOOR AL MAIR

ملاحظات المهندس :



Ali Ahmmed Al Shareef

GOVT. OF RAS AL KHAIMAH - RAK MUNICIPALITY
TOWN PLANNING & SURVEY ADMINISTRATION
حكومة رأس الخيمة - دائرة البلدية
إدارة التخطيط والمساحة

PHOTO OF SITE BH 1

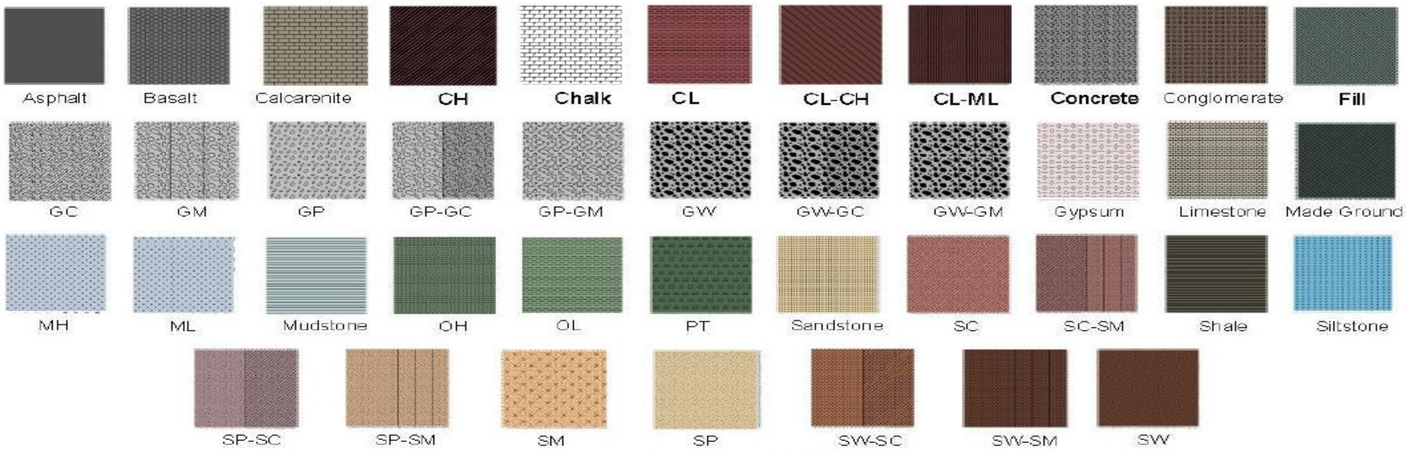
PHOTO OF SITE BH 2

APPENDIX B

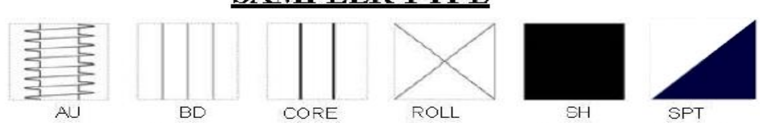
Key to Boreholes
Borehole Logs

LEGEND TO BORING LOGS

SYMBOLS FOR SOIL AND ROCK



SAMPLER TYPE



PENETRATION RESISTANCE AND SOIL PROPERTIES ON BASIS OF THE STANDARD PENETRATION TEST

Fine Grained Soils

S.P.T (Blows/ 30cm)	Consistency	Field Identification	Unconfined Compressive Strenght (Kg/cm^2)
0 – 2	Very Soft	Easily penetrated several cms with fist.	<0.25
2 – 4	Soft	Easily penetrated several cms with thumb.	0.25 – 0.5
4 – 8	Firm	Penetrated several cms by thumb with moderate effort	0.5 – 1.0
8 – 15	Stiff	Readily indented by thumb but penetrated only with great effort.	1.0 – 2.0
15 – 30	Very Stiff	Readily penetrated by thumb nail.	2.0 – 4.0
≥30	Hard	Indented with difficulty by thumb nail.	>4.0

Course Grained Soils

S.P.T (Blows/30cm)	Consistency	Field Identification	Relative Density (%)
0 – 4	Very Loose	Easily Indented with finger, Thumb or fist.	0 – 20
4 – 10	Loose	Less easily indented with fist by easily shoveled	20 – 40
10 – 30	Medium Dense	Shoveled with difficulty	40 – 60
30 – 50	Dense	Requires pick to loosen for shoveling by hand	60 – 80
>50	Very Dense	Requires blasting or heavy equipment to loosen	80 - 100

DEFINITIONS

N-VALUE: The number of blows necessary to produce a penetration of 30cm (1ft.) by dropping a hammer weighing 65kg (140Lbs.) onto the drilling rods from a height of 76cm (30 inches).

TOTAL CORE RECOVERY (TCR): The percentage ratio between the length of core recovered and the total length of core drilled in a given run.

ROCK QUALITY DESIGNATED (RQD): The percentage ratio between the total length of core pieces that are least 10cm (4in.) long and are hard and sound and the length of core drilled in a given run

Point Load Strength and Unconfined Compressive Strength

In the point load test, a rock core is loaded between two steel cones and failures occurs by tensile splitting. A point load strength ind es, I, is calculated as the ratio of the applied load P, at rupture to the square of the distance, H between the loading $I_s = P/H^2$. A correlation that is commonly used between the point index and the unconfined compressive strength. qu of a cylinder with a length to diameter ratio 2 to 1 is qu = $24I_{s(50)}$

Rock

Rock Quality Designation (RQD) (%)	Rock Quality Description
0 – 25	Very poor
25 – 50	Poor
50 – 70	Fair
70 – 90	Good
90 - 100	Excellent

Rock

Description	Field Identification	Unconfined Compressive Strength (Mpa)
Extremely Weak	Can be indented by thumbnail.	0.6 – 1.0
Very Weak	Can be peeled by pocket knife.	1 – 5
Weak	Can be peeled by pocket knife with difficulty.	5 – 25
Medium Strong	Cannot be scrape with pocket knife.	25 – 50
Strong	Requires more than one blow of geological hammer to fracture.	50 – 100
Very Strong	Requires many blows of geological hammer to fracture.	100 – 250
Extremely Strong	Can only be chipped with geological hammer	>250

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CONSULT SOIL TESTING LAB

Testing - NAL 225

BOREHOLE LOG No.:01

Client	:MR. AYMEN BIN HAMADI ESKANDARANI.
Project No	: CST-RAK-2026-9922
Project Name	: (G+1 VILLA)
Location	: Plot#309010298, KHOR AL MUAIREED, RAK- UAE.
Equipment Type	: Rotary
Drilling Fluid Used	: Water
Coordinates	25.825205,55.970771

Log Sheet	: Sheet 1 of 1
Borehole Elevation	
Borehole Dia.	: 150mm
Casing Dia.	: 150mm
Type of Boring	: Rotary
Date Started	: 14/01/2026
Date Finished	: 14/01/2026

DEPTH (m.)	SAMPLE No	SAMPLE Type	S.P.T. No. of Blows 150mm	150mm	150mm	'N'	TCR (%)	SCR (%)	RQD (%)	FI	DESCRIPTION OF STRATA	LEGEND	Layer Thick. (m.)
0.0	1		BULK SAMPLE										0.50
0.5	2	S	9	14	8	22					Medium dense to dense, brown to light brown, slightly silty to silty, fine to medium grained sandy GRAVELS, with BOULDERS, traces of shell		1.50
1.0	3	S	9	13	12	25							
1.5	4	S	12	14	18	32							
2.0	5	S	3	5	4	9					Loose to medium dense, light brown to light grey, slightly silty to silty, fine to medium grained sandy GRAVELS, with BOULDERS, traces of shell		2.50
2.5	6	S	4	6	7	13							
3.0	7	S	6	11	6	17							
3.5													3.40
4.0	8	S	5	3	4	7							
4.5											Medium dense to dense, light grey to greyish , slightly silty to silty, fine to medium grained SAND, with shell fragments.		3.50
5.0	9	S	5	8	7	15							
5.5													
6.0	10	S	6	11	13	24							
6.5													
7.0	11	S	6	13	15	28							
7.5													
8.0	12	S	7	14	18	32					END OF BOREHOLE AT 8m DEPTH		
8.5													

Notes:- Groundwater was encountered below 3.40m From working level , Also at the time of Drilling, working area of site was approximately (-0.50m) down with the ER.LVL.

Ground Water Level : 3.40m below working level.

Logged By: HM

Checked By: DM

Key
TCR Total Core Recovery
SCR Solid Core Recovery
RQD Rock Quality Designation
S.P.T. Standard Penetration Test
FI Fracture Index

S - S.P.T. Sample
B - Bulk Sample
D - Disturbed Sample
C - Coring Sample

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CONSULT SOIL TESTING LAB

Testing - NAL 225

Client	:MR. AYMEN BIN HAMADI ESKANDARANI.
Project No	: CST-RAK-2026-9922
Project Name	: (G+1 VILLA)
Location	: Plot#309010298, KHOR AL MUAIREED, RAK- UAE.
Equipment Type	: Rotary
Drilling Fluid Used	: Water
Coordinates	25.825205,55.970771

BOREHOLE LOG No.:02

Log Sheet	: Sheet 1 of 1
Borehole Elevation	
Borehole Dia.	: 150mm
Casing Dia.	: 150mm
Type of Boring	: Rotary
Date Started	: 14/01/2026
Date Finished	: 14/01/2026

DEPTH (m.)	SAMPLE No	SAMPLE Type	S.P.T. No. of Blows 150mm	150mm	150mm	'N'	TCR (%)	SCR (%)	RQD (%)	FI	DESCRIPTION OF STRATA	LEGEND	Layer Thick. (m.)
_0.0	1		BULK SAMPLE										0.50
_0.5	2	S	7	11	12	23					Medium dense to dense, brown to light brown, slightly silty to silty, fine to medium grained sandy GRAVELS, with BOULDERS, traces of shell		1.50
_1.0	3	S	9	12	13	25							
_1.5	4	S	11	14	17	31							
_2.0	5	S	3	4	4	8					Loose to medium dense, light brown to light grey, slightly silty to silty, fine to medium grained sandy GRAVELS, with BOULDERS, traces of shell		2.50
_2.5	6	S	3	5	6	11							
_3.0	7	S	5	9	7	16							
_3.5													3.40
_4.0	8	S	6	4	5	9							
_4.5											Medium dense to dense, light grey to greyish , slightly silty to silty, fine to medium grained SAND, with shell fragments.		3.50
_5.0	9	S	6	8	9	17							
_5.5													
_6.0	10	S	7	10	13	23							
_6.5													
_7.0	11	S	7	13	16	29							
_7.5													
_8.0	12	S	8	15	19	34					END OF BOREHOLE AT 8m DEPTH		
_8.5													

Notes:- Groundwater was encountered below 3.40m From working level , Also at the time of Drilling, working area of site was approximately (-0.50m) down with the ER.LVL.

Ground Water Level : 3.40m below working level.

Logged By: HM

Checked By: DM

Key

TCR Total Core Recovery

SCR Solid Core Recovery

RQD Rock Quality Designation

S.P.T. Standard Penetration Test

FI Fracture Index

S - S.P.T. Sample

B - Bulk Sample

D - Disturbed Sample

C - Coring Sample

APPENDIX C

Particle Size Distribution Curves

RAK- U.A.E
TEL: 072331646
MOBILE: 0503031180, 0527096722
FAX: 072338860
P.O BOX 33958, RAK

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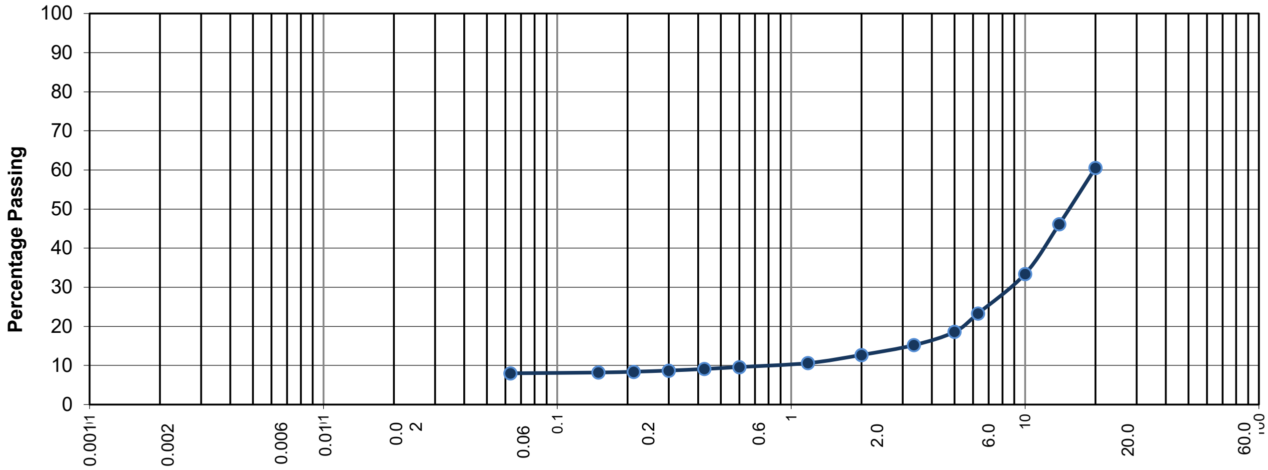
CONSULT SOIL TESTING LAB

Testing - NAL 225

TEST REPORT - PARTICLE SIZE DISTRIBUTION

Client/consult	MR. AYMEN BIN HAMADI ESKANDARANI.	Job Order No.	CST-RAK-2026-9922
Project :	PROPOSED (G+1 VILLA)	Report No :	CST-RAK-2026-9922
Location:	309010298, KHOR AL MUAIREED,RAK, U.A.E	Report Date :	15/01/2026



CLAY	FINE	MEDIUM	COARSE	FINE	MEDIUM	COARSE	FINE	MEDIUM	COARSE	Cobble
	SILT			SAND			GRAVEL			

Borehole No.	Sample No	Depth (m)	Description	Silt/Clay (%)	Sand (%)	Gravel (%)
1	CST-RAK-2026-9922	1.5	slightly silty to silty sandy GRAVELS, BOULDERS	8	5	87

Remarks : None

Ras Al Khaimah U.A.E.
CONSULT SOIL TESTING LABORATORY
مختبر كونسلت لفحص التربة

Authorized Signatory

RAK- U.A.E
TEL: 072331646
MOBILE: 0503031180, 0527096722
FAX: 072338860
P.O BOX 33958, RAK

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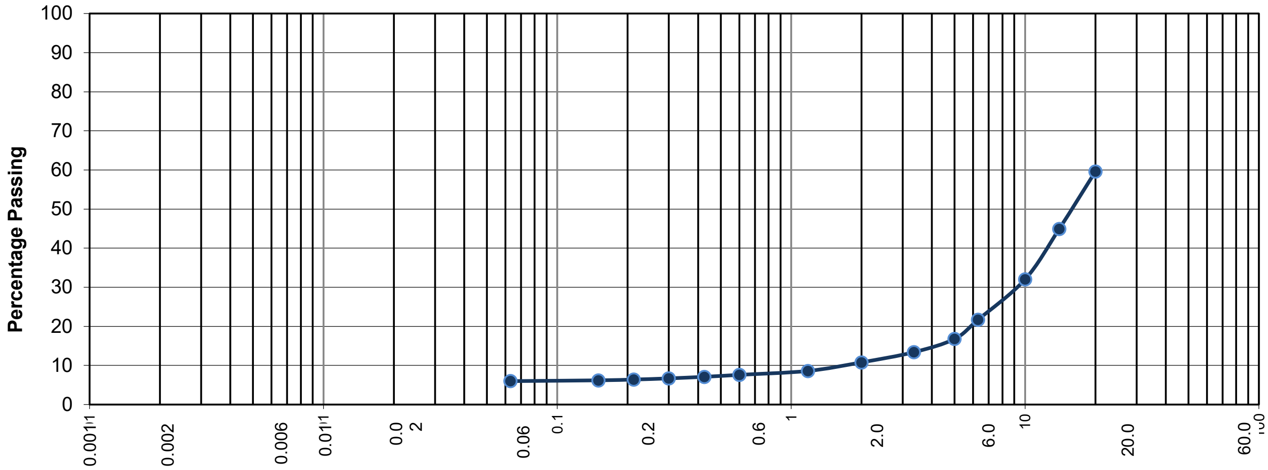
CONSULT SOIL TESTING LAB

EIAS Emirates National Accreditation System
Testing - NAL 225

TEST REPORT - PARTICLE SIZE DISTRIBUTION

Client/consult	MR. AYMEN BIN HAMADI ESKANDARANI.	Job Order No.	CST-RAK-2026-9922
Project :	PROPOSED (G+1 VILLA)	Report No :	CST-RAK-2026-9922
Location:	309010298, KHOR AL MUAIREED,RAK, U.A.E	Report Date :	15/01/2026



CLAY	FINE	MEDIUM	COARSE	FINE	MEDIUM	COARSE	FINE	MEDIUM	COARSE	Cobble
	SILT			SAND			GRAVEL			

Borehole No.	Sample No	Depth (m)	Description	Silt/Clay (%)	Sand (%)	Gravel (%)
1	CST-RAK-2026-9922	2.5	slightly silty to silty sandy GRAVELS, BOULDERS	6	5	89

Remarks : None

Authorized Signatory

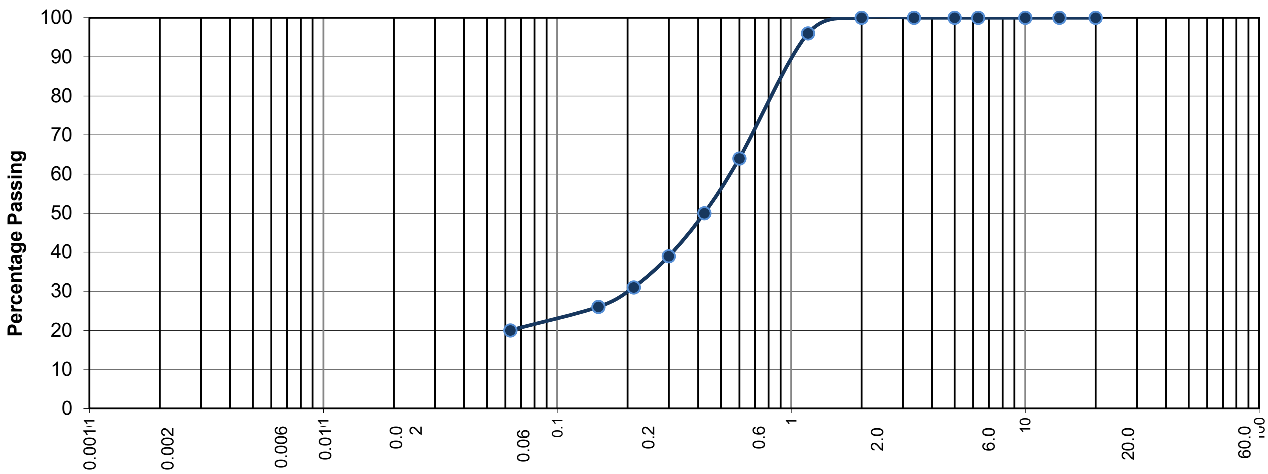
RAK- U.A.E
TEL: 072331646
MOBILE: 0503031180, 0527096722
FAX: 072338860
P.O BOX 33958, RAK

مختبر كونسلت لفحص التربة
CONSULT SOIL TESTING LAB

ENAS Emirates National Accreditation System
Testing - NAL 225

TEST REPORT - PARTICLE SIZE DISTRIBUTION

Client/consult	MR. AYMEN BIN HAMADI ESKANDARANI.	Job Order No.	CST-RAK-2026-9922
Project :	PROPOSED (G+1 VILLA)	Report No :	CST-RAK-2026-9922
Location:	309010298, KHOR AL MUAIREED,RAK, U.A.E	Report Date :	15/01/2026



CLAY	FINE	MEDIUM	COARSE	FINE	MEDIUM	COARSE	FINE	MEDIUM	COARSE	Cobble
	SILT			SAND			GRAVEL			

Borehole No.	Sample No	Depth (m)	Description	Silt/Clay (%)	Sand (%)	Gravel (%)
2	CST-RAK-2026-9922	5	slightly silty to silty SAND, with shell fragments	20	80	0

Remarks : None

Authorized Signatory

APPENDIX D

Chemical Analysis

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CONSULT SOIL TESTING LAB

Ref. No.: **CST-RAK-2026-9922**
Location: **309010298, KHOR AL MUAIREEDH, RAK, U.A.E.**
Report Date: **18/ 01 /2026**

Testing - NAL 225

Chemical Analysis Tests Results

BH No.	Depth (m)	Sulphate Content: Ground Water (g/l) (Nearest 10)	Sulphate Content: 2:1 Soil Extract (Mg/l) (Nearest 10)	Sulphate Content: Soil Acid Extract (%) (Nearest 10)	Sulphate Content: 2:1 pH of the Water Extract (Nearest 0.1)	Chloride Content Cl: Ground water (%) (Nearest 0.01)	Chloride Content Cl: 2:1 Soil Extract (Mg/l) (Nearest 0.01)	Chloride Content Cl: Soil Acid Extract (%) (Nearest 10)	pH Value @ 25ºC (Nearest 0.1)
		SO_4	SO_4	SO_4		Cl	Cl	Cl	
(1)	SOIL (5.00m)		2058		8.07	---	900		8.13
	WATER (3.40m)	2.84	----	----	----	0.07			7.88
(2)	% Passing 2mm sieve =72 (nearest 1.0)								
	Monosulfide Minerals				PRESENT ☐			NOTPRESENT √ ☐	

According to BH #1: The chloride content of the SOIL is Significant.
According to BH #1: The sulphate content of the SOIL is Significant.

Test Methods:

Type of Determination:	***Standards***
Sample Preparation:	
Sample Preparation	BS 1377: Part 1: 2016
Sample Preparation Sulfur (General)	BS 1377: Part 3: 2018+A1:2021: Cl.7.2.3
Preparation of Water: Soil Extract	BS 1377: Part 3: 2018+A1:2021: Cl.7.3.2, 7.3.3
Preparation of Ground Water for Testing	BS 1377: Part 3: 2018+A1:2021: Cl.7.8.2
Preparation of Acid: Soil Extract	BS 1377: Part 3: 2018+A1:2021: Cl.7.9.2, 7.9.4
Preparation of Water Soluble Chloride Extract	BS 1377: Part 3: 2018+A1:2021: Cl.9.2.3
Preparation of Acid Soluble Chloride Extract	BS 1377: Part 3: 2018+A1:2021: Cl.9.3.3.2
Preparation of Sample (pH)	BS 1377: Part 3: 2018+A1:2021: Cl.12.4
Testing Procedures:	
Water Soluble Sulphate Content of Soil (SO4) analysis	BS 1377: Part 3: 2018+A1:2021: Cl.7.3 & 7.6- Gravimetric Method of analysis
Acid Soluble Sulphate Content of Soil (SO4) analysis	BS 1377: Part 3: 2018+A1:2021: Cl.7.9.3 & 7.6 - Gravimetric Method of analysis
Sulphate Content of Groundwater (SO4) Analysis	BS 1377: Part 3: 2018+A1:2021: Cl 7.8 & 7.6 – Gravimetric Method of Analysis
Water Soluble Chloride Content of Soil	BS 1377: Part 3: 2018+A1:2021: Cl.9.2.8.2 - Volhard Method
Acid Soluble Chloride Content of Soil	BS 1377: Part 3: 2018+A1:2021: Cl.9.3.3.3 – Volhard Method
Chloride Content of Groundwater (Cl)	BS 1377: Part 3: 2018+A1:2021: Cl.9.2.7, 9.2.8.2 - Volhard Method
pH Value of Groundwater	BS 1377: Part 3: 2018+A1:2021: Cl. 12.5
pH Value of Soil	BS 1377: Part 3: 2018+A1:2021: Cl. 12.5

APPENDIX E

MOISTURE CONTENT

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TEL: 072331646
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راس الخيمه-الامارات العربيه المتحده
تلفون: 072331646
فاكس:072338860
ص. ب:33958

REPORT ON MOISTURE CONTENT OF SOIL

Contractor	: NIL			
Address	: NIL	Report No.	:	**CST-RAK-2026-9922**
Consultant	: SAVANA	Lab. Sample No	:	CST-RAK-2026-9922
Client name	: MR. AYMAN ESKANDARANI.	Lab. Project No	:	**NP**
Project name	: G+1 VILLA	Sample Certificate	:	NP
Project No.	: CST-RAK-2026-9922	Sample Size (Kg)	:	0.568
Project Location	: KHOR AL MUAIREED,RAK-UAE	Lot No.	:	NP
Sample Description	: SLIGHTLY SILTY TO SILTY SANDY GRAVELS, shell fragments	Lot Size	:	NP
Sample Identification	: NP	Sample brought by	:	Laboratory
Nominal Size	: NP	Date received	:	14/01/2026
Source	: BH-1-1.50M	Date test Started	:	15/01/2026
Sender's Ref.	: NP	Date Test Completed	:	15/01/2026
Sampling Method	: BS EN ISO: 17892-1:2014	Report Date	:	15/01/2026
Sampling date/time	: 14/01/2026	Tested by	:	RE
Sampled by	: CST			

Test Data

Item No.	Test Name	Test Standard	Test Result
1	Moisture Content (%)	BS 1377-2:2022, Cl. 4.1/BS EN 17892-1:2014,A1:2022	**11.26**

Test method variation : None
Remarks : None

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CONSULT SOIL TESTING LABORATORY

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تلفون: 072331646
فاكس:072338860
ص. ب:33958

REPORT ON MOISTURE CONTENT OF SOIL

Contractor	: NIL		
Address	: NIL	Report No.	: **CST-RAK-2026-9922**
Consultant	: SAVANA	Lab. Sample No	: CST-RAK-2026-9922
Client name	: MR. AYMAN ESKANDARANI.	Lab. Project No	: **NP**
Project name	: G+1 VILLA	Sample Certificate	: NP
Project No.	: CST-RAK-2026-9922	Sample Size (Kg)	: 0.568
Project Location	: KHOR AL MUAIREED,RAK-UAE	Lot No.	: NP
Sample Description	: SLIGHTLY SILTY TO SILTY SANDY GRAVELS, shell fragments	Lot Size	: NP
Sample Identification	: NP	Sample brought by	: Laboratory
Nominal Size	: NP	Date received	: 14/01/2026
Source	: BH-1-2.50M	Date test Started	: 15/01/2026
Sender's Ref.	: NP	Date Test Completed	: 15/01/2026
Sampling Method	: BS EN ISO: 17892-1:2014	Report Date	: 15/01/2026
Sampling date/time	: 14/01/2026	Tested by	: RE
Sampled by	: CST		

Test Data

Item No.	Test Name	Test Standard	Test Result
1	Moisture Content (%)	BS 1377-2:2022, Cl. 4.1/BS EN 17892-1:2014,A1:2022	**10.35**

Test method variation : None
Remarks : None

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تلفون: 072331646
فاكس:072338860
ص. ب:33958

REPORT ON MOISTURE CONTENT OF SOIL

Contractor	: NIL		
Address	: NIL	Report No.	: **CST-RAK-2026-9922**
Consultant	: SAVANA	Lab. Sample No	: CST-RAK-2026-9922
Client name	: MR. AYMAN ESKANDARANI.	Lab. Project No	: **NP**
Project name	: G+1 VILLA	Sample Certificate	: NP
Project No.	: CST-RAK-2026-9922	Sample Size (Kg)	: 0.568
Project Location	: KHOR AL MUAIREED,RAK-UAE	Lot No.	: NP
Sample Description	: SLIGHTLY SILTY TO SILTY SANDY GRAVELS, shell fragments	Lot Size	: NP
Sample Identification	: NP	Sample brought by	: Laboratory
Nominal Size	: NP	Date received	: 14/01/2026
Source	: BH-2-5.0M	Date test Started	: 15/01/2026
Sender's Ref.	: NP	Date Test Completed	: 15/01/2026
Sampling Method	: BS EN ISO: 17892-1:2014	Report Date	: 15/01/2026
Sampling date/time	: 14/01/2026	Tested by	: RE
Sampled by	: CST		

Test Data

Item No.	Test Name	Test Standard	Test Result
1	Moisture Content (%)	BS 1377-2:2022, Cl. 4.1/BS EN 17892-1:2014,A1:2022	**17.45**

Test method variation : None
Remarks : None

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APPENDIX F

* REQUIREMENTS OF DIFFERENT SOURCES FOR CONCRETE EXPOSED TO SULPHATE ATTACK

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CONSULT SOIL TESTING LAB

TEST REPORT

SPECIFIC GRAVITY

ISO 17892-3:2015

Client **MR. AYMAN ESKANDARANI.**

Report Date : 17/01/2026

Sample Number	: CST-RAK-2026-9922	**Report Number**	: CST-RAK-2026-9922
Project Number	: CST-RAK-2026-9922	**Plot No.**	: 309010298
Project Name	: G+1 VILLA	**Sample Source**	: 5.00 M
Contractor	: NP	**Sample Location**	: BH-1
Project Client	: MR. AYMAN ESKANDARANI.	**Sampled By**	: CST
Project Location	: PLOT:309010298, KHOR ALMUAIREED, RAK-UAE	**Sampling Date/Time**	: 15/01/2026
Consultant	: NP	**Sampling Method**	: ISO 17892-3
Client Ref./Req. No.	:	**Sample Delivered By**	: CST
Sample Size	: 0.450 KG	**Sample Received Date/Time**	: 15/01/2026

Test Results :

Specific Gravity
2.70

Sample Description : Slightly Silty Sand

Testing Date : **17/01/2026**

Tested by : MN

Certificate of Sampling : **Not given**

Sample preparation metho : **ISO 17892-3:2015**

Test Specimen Preparation : **ISO 17892-3:2015**

Test Method : **ISO 17892-3:2015**

Test variation : :

Remarks : :

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Table A.1 *(continued)*

Class designation	Class description	Informative examples applicable in the United Kingdom

A) The moisture condition relates to that in the concrete cover to reinforcement or other embedded metal but, in many cases, conditions in the concrete cover can be taken as being that of the surrounding environment. This might not be the case if there is a barrier between the concrete and its environment (see A.3).

B) [A1⟩ Soil may be classed as not containing chlorides if the chloride level is not greater than 275 mg/l. Higher limits might be applicable; seek specialist advice. ⟨A1]

C) [A1⟩ Where the integrity of the waterproofing over the intended working life of the concrete surface it protects cannot be guaranteed or maintained then it may be prudent to assume XD1 or XS1 exposure class as appropriate. ⟨A1]

D) Reinforced and prestressed concrete elements where one surface is immersed in water containing chlorides and another is exposed to air are potentially in a more severe exposure condition, especially where the dry side is at a high ambient temperature. Specialist advice should be sought where appropriate, to develop a specification that is appropriate to the actual conditions likely to be encountered.

E) In deeply buried or fully submerged conditions, the availability of oxygen might be significantly restricted and consequently the risk of damage due to corrosion of reinforcement could be much reduced.

F) See A.4.3 and A.4.4 as reinforced pavements and car park slabs might be needed to resist freeze-thaw attack.

G) The rate of ingress of chloride into the concrete depends on the concentration at its surface: brackish groundwater (chloride content less than 18 g/l) is less severe than exposure to sea water.

H) Exposure XS3 covers a range of conditions. The most extreme conditions are in the spray zone. The least extreme is in the tidal zone where conditions can be similar to those in XS2. The recommendations given in this Annex take into account the most extreme conditions within this class.

I) It is not normally necessary to classify the XF4 exposure classes to those parts of structures located in the United Kingdom which are in frequent contact with the sea.

Table A.2 — *Aggressive chemical environment for concrete (ACEC) exposure classes*

Sulfate and magnesium					Design sulfate class	Natural soil		Brownfield[A)]		ACEC-class
2:1 water/soil extract[B)]		Groundwater[B)]		Total potential sulfate[C)]		Static water	Mobile water	Static water	Mobile water	
SO_4[D)] mg/l	Mg[E) D)] mg/l	SO_4[D)] mg/l	Mg[E) D)] mg/l	SO_4 %		pH	pH	pH[F)]	pH[F)]	
						≥2.5	—	≥2.5	—	AC-1s
						—	>5.5	—	>6.5	AC-1[G)]
<400	—	≤300	—	<0.24	DS-1	—	2.5 to 5.5	—	5.6 to 6.5	AC-2z
						—	—	—	4.5 to 5.5	AC-3z
						—	—	—	2.5 to 4.5	AC-4z
						>3.5	—	>5.5	—	AC-1s

Table A.2 *(continued)*

Sulfate and magnesium					Design sulfate class	Natural soil		Brownfield[A)]		ACEC-class
2:1 water/soil extract[B)]		Groundwater[B)]		Total potential sulfate[C)]		Static water	Mobile water	Static water	Mobile water	
SO_4[D)] mg/l	Mg[E) D)] mg/l	SO_4[D)] mg/l	Mg[E) D)] mg/l	SO_4 %		pH	pH	pH[F)]	pH[F)]	
						—	>5.5	—	>6.5	AC-2
500 to 1 500	—	400 to 1 400	—	0.24 to 0.6	DS-2	2.5 to 3.5	—	2.5 to 5.5	—	AC-2s
						—	2.5 to 5.5	—	5.6 to 6.5	AC-3z
						—	—	—	4.5 to 5.5	AC-4z
						—	—	—	2.5 to 4.5	AC-5z
						>3.5	—	>5.5	—	AC-2s
						—	>5.5	—	>6.5	AC-3
1 600 to 3 000	—	1 500 to 3 000	—	0.7 to 1.2	DS-3	2.5 to 3.5	—	2.5 to 5.5	—	AC-3s
						—	2.5 to 5.5	—	5.6 to 6.5	AC-4
						—	—	—	2.5 to 5.5	AC-5
						>3.5		>5.5	—	AC-3s
3 100 to 6 000	≤1 200	3 100 to 6 000	≤1 000	1.3 to 2.4	DS-4	—	>5.5	—	>6.5	AC-4
						2.5 to 3.5	—	2.5 to 5.5	—	AC-4s
						—	2.5 to 5.5	—	2.5 to 6.5	AC-5
								>5.5		AC-3s
3 100 to 6 000	>1 200 [E)]	3 100 to 6 000	>1 000 [E)]	1.3 to 2.4	DS-4m	Not found in UK natural ground			>6.5	AC-4m
								2.5 to 5.5		AC-4ms
									2.5 to 6.5	AC-5m
>6 000	≤1 200	>6 000	≤1 100	>2.4	DS-5	>3.5	—	>5.5	—	AC-4s
						2.5 to 3.5	W2.5	2.5 to 5.5	W2.5	AC-5
>6 000	>1 200 [E)]	>6 000	>1 100 [E)]	>2.4	DS-5m	Not found in UK natural ground		>5.5	—	AC-4ms
								2.5 to 5.5	W2.5	AC-5m

Table A.2 *(continued)*

Sulfate and magnesium					Design sulfate class	Natural soil		Brownfield A)		ACEC-class
2:1 water/soil extract B)		Groundwater B)		Total potential sulfate C)		Static water	Mobile water	Static water	Mobile water	
SO_4 D) mg/l	Mg E) D) mg/l	SO_4 D) mg/l	Mg E) D) mg/l	SO_4 %		pH	pH	pH F)	pH F)	

A) "Brownfield" sites are those that might contain chemical residues remaining from previous industrial use or from imported wastes.

B) Characteristic values determined in accordance with BRE Special Digest 1 [1].

C) Applies only to sites where concrete is exposed to sulfate ions (SO_4), which can result from the oxidation of sulfides such as pyrite, following ground disturbance.

D) Values should be rounded to 100 mg/l.

E) The limit on water-soluble magnesium does not apply to brackish groundwater (chloride content between 12 g/l and 18 g/l). This allows these sites to be classified in the row above. This Table does not cover sea water (see **A.4.6** and Table A.13) and stronger brines.

F) An additional account is taken of hydrochloric and nitric acids by adjustment to sulfate content (see BRE Special Digest 1 [1]).

G) For flowing water that is potentially aggressive to concrete owing to high purity or an aggressive carbon dioxide level greater than 15 mg/l, increase the ACEC class to AC-2z (see BRE Special Digest 1 [1]).

A.2.2 Environments related to corrosion of reinforcement

There are four environments related to corrosion of reinforcement: X0, XC, XD and XS (see Table A.1). If the concrete is reinforced or contains embedded metal, one of these exposure classes should be identified. Where concrete contains reinforcement, it is recommended that the X0 exposure class is only used where the relative humidity is very low (less than about 35%) and this condition is rarely found in practice. In most situations, exposure class XC1 exists on its own. Where the concrete is classified as being exposed to environmental class XC1 on the basis of it being permanently wet, XC1 may be combined with freeze-thaw exposure class XF3. Exposure class XC2 can exist on its own in a buried foundation in soil classed as AC-1 or AC-1s where the hydraulic gradient is not greater than 5 and beyond the zone where freeze-thaw attack is a consideration. It can also exist in combination with exposure class XF3 and/or one of the chemically aggressive exposure classes.

Exposure class XC3 (moderate humidity) has been combined with XC4 (cyclic wet and dry), because the recommendations for the concrete specification and cover to reinforcement are the same for XC3 and XC4.

A.2.3 Environments associated with unreinforced concrete

The classification of exposure for unreinforced concrete is limited to exposure classes X0, ACEC, XAS and/or XF. The XC, XD and XS classes are not applicable as they relate specifically to the risk of corrosion of reinforcement.

Exposure class X0 can exist only on its own. It is not normally necessary to classify in the XF4 exposure class those parts of structures located in the United Kingdom which are in frequent contact with the sea as is the case for XAS exposure. An aggressive chemical environment for concrete (ACEC class) can apply on its own or in combination with an XF exposure class. If the unreinforced concrete contains any embedded metal,it should be classified as reinforced and the appropriate limiting values associated with exposure classes XC, XD or XS should be selected.

Table A.6 — *Cement and combination types*[A)]

Broad designation[B)]	**Composition**	**Comprises cement and combination types (see BS 8500-2:2015+[A2⟩ A2:2019 ⟨A2], Table 1)**
CEM I	Portland cement	CEM I
CEM I-SR 0 CEM I-SR 3	Sulfate-resisting Portland cement	CEM I-SR 0 CEM I-SR 3
IIA	[A2⟩ Portland cement with either: 6% to 20% pozzolana, 6% to 20% fly ash, 6% to 20% ground granulated blastfurnace slag, 6% to 20% limestone, or 6% to 10% silica fume [C)]. Portland cement with either: 6% to 20% limestone and pozzolana, or 6% to 20% limestone and ground granulated blastfurnace slag, or 6% to 20% limestone and fly ash ⟨A2]	CEM II/A-L [A2⟩ or LL ⟨A2], CIIA-L [A2⟩ or LL ⟨A2], CEM II/A-S, CIIA-S, [A2⟩ CEM II/A-P or Q, CIIA-P or Q, ⟨A2] CEM II/A-V, CIIA-V, CEM II/A-D[A2⟩ , CEM II/A-M, CIIA-M ⟨A2]
IIB-S	Portland cement with 21% to 35% ground granulated blastfurnace slag	CEM II/B-S, CIIB-S
[A2⟩ IIB-P or Q	Portland cement with 21% to 35% pozzolana	CEM II/B-P or Q, CIIB-P or Q ⟨A2]
IIB-V	Portland cement with 21% to 35% fly ash	CEM II/B-V, CIIB-V
[A2⟩ IIB-M	Portland composite cement, comprised not less than 65% Portland cement clinker with either: 21% to 35% pozzolana and limestone [D)], 21% to 35% fly ash and limestone [D)], or 21% to 35% ground granulated blastfurnace slag and limestone [D)]. Portland limestone cement with up to either 29% pozzolana, 29% fly ash, or 29% ground granulated blastfurnace slag with not less than 65% Portland cement clinker in the combination.	CEM II/B-M(P or Q-L or LL), CIIB-M(P or Q-L or LL) CEM II/B-M (V-L or LL), CIIB-M (V-L or LL) CEM II/B-M (S-L or LL), CIIB-M (S-L or LL) CEM II/B-M(L or LL-P or Q), CIIB-M(L or LL-P or Q) CEM II/B-M(L or LL-V), CIIB-M(L or LL-V) CEM II/B-M(L or LL-S), CIIB-M(L or LL-S) ⟨A2]
[A2⟩ *Text deleted*	*Text deleted*	*Text deleted* ⟨A2]
[A2⟩ IIB-P+SR IIB-Q+SR	Portland cement with 25% to 35% pozzolana	CEM II/B-P+SR, CIIB-P+SR CEM II/B-Q+SR, CIIB-Q+SR ⟨A2]
[A2⟩ IIB-V+SR	Portland cement with 25% to 35% fly ash	CEM II/B-V+SR, CIIB-V+SR ⟨A2]

Table A.6 *(continued)*

Broad designation[B)]	**Composition**	**Comprises cement and combination types (see BS 8500-2:2015+[A2⟩ A2:2019 ⟨A2], Table 1)**
IIIA[A2⟩ E) ⟨A2]	Portland cement with 36% to 65% ground granulated blastfurnace slag	CEM III/A, CIIIA
IIIA+SR	Portland cement with 36% to 65% ground granulated blastfurnace slag with additional requirements that enhance sulfate resistance	CEM III/A+SR[A2⟩ F) ⟨A2], CIIIA+SR [A2⟩ F) ⟨A2]
IIIB[A2⟩ G) ⟨A2]	Portland cement with 66% to 80% ground granulated blastfurnace slag	CEM III/B, CIIIB
IIIB+SR	Portland cement with 66% to 80% ground granulated blastfurnace slag with additional requirements that enhance sulfate resistance	CEM III/B+SR[A2⟩ F) ⟨A2], CIIIB+SR[A2⟩ F) ⟨A2]
[A2⟩ IVB-P H) IVB-Q H)	Portland cement with 36% to 55% pozzolana	CEM IV/B-P, CIVB-P CEM IV/B-Q, CIVB-Q ⟨A2]
IVB-V[A2⟩ I) ⟨A2]	Portland cement with 36% to 55% fly ash	CEM IV/B-V, CIVB-V

A) There are a number of cements and combinations not listed in this Table that may be specified for certain specialist applications. See BRE Special Digest 1 [1] for the sulfate-resisting characteristics of other cements and combinations. See IP 17/05 [5] for the use of high ggbs content cements and combinations in secant piling applications.

B) The use of these broad designations is sufficient for most applications. Where a more limited range of cement or combinations types is required, select from the notations given in BS 8500-2:2015+[A2⟩ A2:2019 ⟨A2], Table 1.

C) When IIA or IIA-D is specified, CEM I and silica fume may be combined in the concrete mixer using the k-value concept; see BS EN 206:2013[A2⟩ +A1:2016 ⟨A2], **5.2.5.2.3**.

D) [A2⟩ Maximum limestone content 20%.

E) Where IIIA is specified, IIIA+SR may be used.

F) "+SR" indicates additional restrictions on the chemical composition of cement or ggbs related to sulfate resistance. See BS 8500-2:2015+A2:2019, Table 1, footnote G.

G) Where IIIB is specified, IIIB+SR may be used.

H) IVA cements and combinations with pozzolana should be classified as II-P or II-Q.

I) IVA cements and combinations with a siliceous fly ash should be classified as II-V. ⟨A2]

Table A.7 — *Minimum cement and combination contents with maximum aggregate sizes other than 20 mm*

Limiting values given for 20 mm maximum aggregate size		**Maximum aggregate size**		
Maximum w/c ratio	**Minimum cement or combination content kg/m³**	**≥40 mm**	**14 mm**	**10 mm**
0.70	240	240	260	280
0.65	260	240	280	300
0.60	300	280	320	340
0.60	280	260	300	320
0.55	300	280	320	340
	320	300	340	360
0.50	320	300	340	360
	340	320	360	380
0.45	340	320	360	360
	360	340	380	380
0.40	380	360	380	380
0.35	380	380	380	380

The specifications for the RC-series of designated concretes include requirements for maximum w/c ratio and minimum cement or combination content, which makes specification easier and complete. However, for designed concrete, conformity to the recommended compressive strength class does not necessarily ensure conformity to requirements for maximum w/c ratio or minimum cement or combination content. If a maximum w/c ratio or minimum cement content is required, it should be specified. Failure to specify means that there is no requirement, as neither BS EN 206 nor BS 8500-2 have default values for designed concrete. The basic requirements detailed in **4.3** were selected to avoid this potential deficiency.

Models to predict concrete requirements for long-life structures in chloride environments do not give identical predictions, and extrapolating performance from existing structures also has many practical problems. Consequently there is a degree of uncertainty with the recommendations for an intended working life of at least 100 years in the chloride (XD) and sea water (XS) environments. Reliance solely on cover and concrete quality might not be the most economic solution. In these situations, consideration may be given to using other techniques such as stainless steel or non-ferrous reinforcement, barriers, coatings and corrosion inhibitors, but these techniques also have their uncertainties. For guidance on these techniques see specialist literature, e.g. Concrete Society Technical Report 61 [6].

The concrete chloride class (the number in the classification being the maximum chloride ion by mass of cement or combination) needs to be specified for prescribed concrete, and for designated and designed concrete when the default chloride class does not apply.

Table A.8 gives recommended chloride classes for particular concrete uses containing reinforcing steel. No guidance is given for galvanized reinforcing steel, epoxy-coated reinforcing steel, corrosion resistant reinforcing steel or any non-steel reinforcement.

A.4.5 Concrete properties and limiting values to resist chemical attack

From the ACEC classification (see Table A.2), the intended working life and the hydraulic gradient, the quality of concrete, expressed as a design chemical class (DC-class), and additional protective measure (APM) should be selected using Table A.10. The APMs are listed in Table A.11. Table A.10 also gives lowest nominal cover recommended in the National Annex to BS EN 1992-1-1.

BRE Special Digest 1 [1] gives full guidance that aids the selection of the DC-class and the relevant APM. The DC-class or (FND) designated concrete specified to the producer should include any adjustments as the result of applying APM1 (enhanced concrete quality).

Table A.10 — *Selection of the nominal cover and DC-class or designated concrete and APM for in-situ concrete elements[A)] in contact with the ground where the hydraulic gradient due to groundwater is five or less[B)C)D)]*

ACEC-class	Lowest nominal cover[E)], mm	Intended working life[F)]	
		At least 50 years[G) H)]	At least 100 years
AC-1s, AC-1	(25 + Δc)	DC-1 (RC25/30 if reinforced)	DC-1 (RC25/30 if reinforced)
AC-2s, AC-2	(25 + Δc)	DC-2 (FND2)	DC-2 (FND2)
AC-2z	(25 + Δc)	DC-2z (FND2Z)	DC-2z (FND2Z)
AC-3s	(25 + Δc)	DC-3 (FND3)	DC-3 (FND3)
AC-3z	(25 + Δc)	DC-3z (FND3Z)	DC-3z (FND3Z)
AC-3	(25 + Δc)	DC-3 (FND3)	DC-3 + one APM of choice, FND3 + one APM of choice, DC-4 or FND4
AC-4s	(25 + Δc)	DC-4 (FND4)	DC-4 (FND4)
AC-4z	(25 + Δc)	DC-4z (FND4Z)	DC-4z (FND4Z)
AC-4	(25 + Δc)	DC-4 (FND4)	DC-4 + one APM from APM2 to APM5, or FND4 + one APM from APM2 to APM5
AC-4ms	(25 + Δc)	DC-4m (FND4M)	DC-4m (FND4M)
AC-4m	(25 + Δc)	DC-4m (FND4M)	DC-4m + one APM from APM2 to APM5, [A1) or FND4M (A1] + one APM from APM2 to APM5
AC-5z	(25 + Δc)	DC-4z (FND4Z) + APM3[I)]	DC-4z (FND4Z) + APM3[I)]
AC-5	(25 + Δc)	DC-4 (FND4) + APM3[I)]	DC-4 (FND4) + APM3[I)]
AC-5m	(25 + Δc)	DC-4m (FND4M) + APM3[I)]	DC-4m (FND4M) + APM3[I)]

Table 1 — *General purpose cements and combinations*

Type	Notation	Standard	Broad designation	Grouping used in BRE SD1:2005 [2]
Portland cement	CEM I	BS EN 197-1	CEM I	A
[A2⟩ Sulfate-resisting Portland cements	CEM I-SR 0	BS EN 197-1	CEM I-SR 0	G
	CEM I-SR 3	BS EN 197-1	CEM I-SR 3	G ⟨A2]
Portland silica fume cement[A)]	CEM II/A-D	BS EN 197-1	IIA	A
Portland limestone cements	CEM II/A-L	BS EN 197-1	IIA	B[B)] or C[B)]
	CEM II/A-LL	BS EN 197-1	IIA	B[B)] or C[B)]
Portland slag cements	CEM II/A-S	BS EN 197-1	IIA	A
	CEM II/B-S	BS EN 197-1	IIB-S	A
[A2⟩ Portland natural pozzolana cements	CEM II/A-P	BS EN 197-1	IIA	A
	CEM II/B-P	BS EN 197-1	IIB-P	A
	CEM II/B-P+SR[C)]	BS EN 197-1	IIB-P+SR	D ⟨A2]
[A2⟩ Portland natural calcined pozzolana cements	CEM II/A-Q	BS EN 197-1	IIA	A
	CEM II/B-Q	BS EN 197-1	IIB-Q	A
	CEM II/B-Q+SR[D)]	BS EN 197-1	IIB-Q+SR	D ⟨A2]
Portland fly ash cements	CEM II/A-V	BS EN 197-1	IIA	A
	CEM II/B-V	BS EN 197-1	IIB-V	A
	CEM II/ B-V+SR[A2⟩ [E)] ⟨A2]	BS EN 197-1	IIB[A2⟩ -V ⟨A2]+SR	D
[A2⟩ Portland composite cements [F)]	CEM II/A-M (S-L or LL)	BS EN 197-1	IIA	B [B)] or C [B)]
	CEM II/A-M (L or LL -S)	BS EN 197-1	IIA	B [B)] or C [B)]
	CEM II/A-M (P or Q-L or LL)	BS EN 197-1	IIA	B [B)] or C [B)]
	CEM II/A-M (L or LL-P or Q)	BS EN 197-1	IIA	B [B)] or C [B)]
	CEM II/A-M (V-L or LL)	BS EN 197-1	IIA	B [B)] or C [B)]
	CEM II/A-M (L or LL-V)	BS EN 197-1	IIA	B [B)] or C [B)]
	CEM II/B-M (S-L or LL)	BS EN 197-1	IIB-M	B [B)] or C [B)]
	CEM II/B-M (L or LL-S)	BS EN 197-1	IIB-M	B [B)] or C [B)]
	CEM II/B-M (P or Q-L or LL)	BS EN 197-1	IIB-M	B [B)] or C [B)]
	CEM II/B-M (L or LL-P or Q)	BS EN 197-1	IIB-M	B [B)] or C [B)]
	CEM II/B-M (V-L or LL)	BS EN 197-1	IIB-M	B [B)] or C [B)]
	CEM II/B-M (L or LL-V)	BS EN 197-1	IIB-M	B [B)] or C [B)] ⟨A2]

Table 1 *(continued)*

Type	Notation	Standard	Broad designation	Grouping used in BRE SD1:2005 [2]
Blastfurnace cements	CEM III/A	BS EN 197-1	IIIA	A
	CEM III/A+SR[A2> G) <A2]	BS EN 197-1	IIIA+SR	D
	CEM III/B	BS EN 197-1	IIIB	A
	CEM III/B+SR[A2> G) <A2]	BS EN 197-1	IIIB+SR	F
Pozzolanic cement[A2> s <A2]	[A2> CEM IV/B(P) H) I) <A2]	[A2> BS EN 197-1 or BS EN 14216 <A2]	[A2> IVB-P <A2]	[A2> E <A2]
	[A2> CEM IV/B(Q) J) K) <A2]	[A2> BS EN 197-1 or BS EN 14216 <A2]	[A2> IVB-Q <A2]	[A2> E <A2]
	CEM IV/B(V)[A2> L) <A2]	BS EN 197-1 or BS EN 14216	IVB-V	E
[A2> *Text deleted* <A2]	[A2> *Text deleted* <A2]	[A2> *Text deleted* <A2]	[A2> *Text deleted* <A2]	[A2> *Text deleted* <A2]
Combinations conforming to Annex A [A2> are manufactured in the concrete mixer. They may be combinations of Portland cement and natural pozzolana, natural calcined pozzolana, high reactivity natural calcined pozzolana, fly ash, ggbs or limestone fines. They may also be combinations of CEM II/A-L or CEM II/A -LL cement with either natural pozzolana, natural calcined pozzolana, high reactivity natural calcined pozzolana, fly ash or ggbs. <A2]				
[A2> CEM I cement conforming to BS EN 197-1 with a mass fraction of 6% to 20% of combination of limestone fines conforming to BS 7979	CIIA-L CIIA-LL	BS 8500-2, Annex A	IIA IIA	B B) or C B) B B) or C B)
CEM I cement conforming to BS EN 197-1 with a mass fraction of 6% to 20% of combination of ggbs conforming to BS EN 15167-1	CIIA-S	BS 8500-2, Annex A	IIA	A
CEM I cement conforming to BS EN 197-1 with a mass fraction of 21% to 35% of combination of ggbs conforming to BS EN 15167-1	CIIB-S	BS 8500-2, Annex A	IIB-S	A
CEM I cement conforming to BS EN 197-1 with a mass fraction of 6% to 20% of combination of natural pozzolana conforming to BS 8615-1	CIIA-P	BS 8500-2, Annex A	IIA	A
CEM I cement conforming to BS EN 197-1 with a mass fraction of 21% to 35% of combination of natural pozzolana conforming to BS 8615-1	CIIB-P CIIB-P+SR C)	BS 8500-2, Annex A	IIB-P IIB-P+SR	A D

Table 1 *(continued)*

Type	Notation	Standard	Broad designation	Grouping used in BRE SD1:2005 [2]
CEM I cement conforming to BS EN 197-1 with a mass fraction of 6% to 20% of combination of natural calcined pozzolana to BS 8615-1 or high reactivity natural calcined pozzolana conforming to BS 8615-2	CIIA-Q	BS 8500-2, Annex A	IIA	A
CEM I cement conforming to BS EN 197-1 with a mass fraction of 21% to 35% of combination of natural calcined pozzolana to BS 8615-1 or high reactivity natural calcined pozzolana conforming to BS 8615-2	CIIB-Q CIIB-Q+SR D)	BS 8500-2, Annex A	IIB-Q IIB-Q+SR	A D ⟨A2]
CEM I cement conforming to BS EN 197-1 with a mass fraction of 6% to 20% of combination of fly ash conforming to BS EN 450-1	CIIA-V	BS 8500-2, Annex A	IIA	A
CEM I cement conforming to BS EN 197-1 with a mass fraction of 21% to 35% of combination of fly ash conforming to BS EN 450-1	CIIB-V CIIB-V+SR[A2⟩ E) ⟨A2]	BS 8500-2, Annex A	IIB-V IIB[A2⟩ -V ⟨A2]+SR	A D
[A2⟩ CEM II/A-L or LL conforming to BS EN 197-1 with a mass fraction of 6% to 29% of combination of ggbs conforming to BS EN 15167-1, and where the mass fraction of Portland cement clinker of combination is not less than 65%	CIIB-M (S-L or LL) F) CIIB-M (L or LL-S)	BS 8500-2, Annex A	IIB-M	B B) or C B)
CEM II/A-L or LL conforming to BS EN 197-1 with a mass fraction of 6% to 29% of combination of fly ash conforming to BS EN 450-1, and where the mass fraction of Portland cement clinker of combination is not less than 65%	CIIB-M (V-L or LL) F) CIIB-M (L or LL-V)	BS 8500-2, Annex A	IIB-M	B B) or C B)
CEM I cement conforming to BS EN 197-1 with a mass fraction of 36% to 65% of combination of ggbs conforming to BS EN 15167-1	CIIIA CIIIA+SR G)	BS 8500-2, Annex A	IIIA IIIA+SR	A F
CEM I cement conforming to BS EN 197-1 with a mass fraction of 66% to 80% of combination of ggbs conforming to BS EN 15167-1	CIIIB CIIIB+SR G)	BS 8500-2, Annex A	IIIB IIIB+SR	A F

Table 1 *(continued)*

Type	Notation	Standard	Broad designation	Grouping used in BRE SD1:2005 [2]
CEM I cement conforming to BS EN 197-1 with a mass fraction of 36% to 55% of combination of natural pozzolana conforming to BS 8615-1	CIVB-P [I)]	BS 8500-2, Annex A	IVB-P	E
CEM I cement conforming to BS EN 197-1 with a mass fraction of 36% to 55% of combination of natural calcined pozzolana to BS 8615-1 or high reactivity natural calcined pozzolana conforming to BS 8615-2	CIVB-Q [K)]	BS 8500-2, Annex A	IVB-Q	E A2
CEM I cement conforming to BS EN 197-1 with a mass fraction of 36% to 55% of combination of fly ash conforming to BS EN 450-1	CIVB-V	BS 8500-2, Annex A	IVB-V	E
A2 *Text deleted* A2	A2 *Text deleted* A2	A2 *Text deleted* A2	A2 *Text deleted* A2	A2 *Text deleted* A2
A2 *Text deleted* A2	A2 *Text deleted* A2	A2 *Text deleted* A2	A2 *Text deleted* A2	A2 *Text deleted* A2
A2 *Text deleted* A2	A2 *Text deleted* A2	A2 *Text deleted* A2	A2 *Text deleted* A2	A2 *Text deleted* A2
A2 *Text deleted* A2	A2 *Text deleted* A2	A2 *Text deleted* A2	A2 *Text deleted* A2	A2 *Text deleted* A2
A2 *Text deleted* A2	A2 *Text deleted* A2	A2 *Text deleted* A2	A2 *Text deleted* A2	A2 *Text deleted* A2

Table 1 *(continued)*

Type	Notation	Standard	Broad designation	Grouping used in BRE SD1:2005 [2]

A) When IIA or IIA-D is specified, CEM I and silica fume may be combined in the concrete mixer using the k-value concept; see [A1⟩ BS EN 206:2013[A2⟩ +A1:2016 ⟨A2], **5.2.5.2.3** ⟨A1].

B) The classification is B if the cement or combination strength is class 42,5 or higher and C if it is class 32,5.

[A2⟩ C) With a minimum proportion of natural pozzolana of 25%. The performance of CEM II/B-P cement and its equivalent combination CIIB-P are not covered by BRE SD1:2005 [2] but are categorized as 'D' on the basis that the performance of natural pozzolana in concrete is assumed to be similar to fly ash.

D) With a minimum proportion of natural calcined pozzolana or high reactivity natural calcined pozzolana of 25%. The performance of CEM II/B-Q cement and its equivalent combination CIIB-Q are not covered by BRE SD1:2005 [2] but are categorized as 'D' on the basis that the performance of natural calcined pozzolana or high reactivity natural calcined pozzolana in concrete is assumed to be similar to fly ash.

E) With a minimum proportion of fly ash of 25%.

F) Within the brackets the constituent listed first is the constituent with the highest proportion, e.g. (LL-S) means the proportion of limestone is greater than the proportion of ggbs and (V-L) means the proportion of siliceous fly ash is greater than the proportion of limestone.

G) Where the alumina content of the slag exceeds 14%, the tricalcium aluminate content of the Portland cement fraction shall not exceed 10%.

H) CEM IV/A cement with natural pozzolana should be classified as either CEM II/A-P (6%–20% natural pozzolana) or CEM II/B-P (21%–35% natural pozzolana). ⟨A2]

I) The performance of CEM IV/B(P) cement and its equivalent combination IV/B(P) are not covered by BRE SD1:2005 [2] but are categorized as 'E' on the basis that the performance of natural pozzolana in concrete is assumed to be similar to fly ash.

J) CEM IV/A cement with natural calcined pozzolana or high reactivity natural calcined pozzolana should be classified as either CEM II/A-Q (6%–20% pozzolana) or CEM II/B-Q (21%–35% pozzolana).

K) The performance of CEM IV/B(Q) cement and its equivalent combination IV/B(Q) are not covered by BRE SD1:2005 [2] but are categorized as 'E' on the basis that the performance of natural calcined pozzolana or high reactivity natural calcined pozzolana in concrete is assumed to be similar to fly ash.

L) CEM IV/A cement with siliceous fly ash should be classified as either CEM II/A-V (6%–20% siliceous fly ash) or CEM II/B-V (21%–35% siliceous fly ash).

4.3.2 Requirements for aggregates

Where D_{upper} is specified, the producer shall supply concrete with a D_{max} not greater than D_{upper}.

NOTE 1 Where D_{max} is specified, either in error or in accordance with previous editions of this standard, then the producer is likely to assume D_{max} equals D_{upper} and that a D_{lower} has not been specified. In practical terms this means that specifying D_{max} in accordance with a previous version of this standard is the same as specifying D_{max} to this edition of the standard.

Where D_{lower} is specified, the producer shall supply concrete with a D_{max} not less than D_{lower}.

The aggregate drying shrinkage shall be not more than 0.075% when determined in accordance with BS EN 12620, unless otherwise specified.

Special Digest 1:2005

Third edition

Concrete in aggressive ground

BRE Construction Division

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The Concrete Centre™

Table C1 Aggressive Chemical Environment for Concrete (ACEC) classification for natural ground locations[a]

Sulfate				Groundwater		ACEC
Design Sulfate Class for location	2:1 water/soil extract[b]	Groundwater	Total potential sulfate[c]	Static water	Mobile water	Class for location
1	2 (SO_4 mg/l)	3 (SO_4 mg/l)	4 (SO_4 %)	5 (pH)	6 (pH)	7
DS-1	< 500	< 400	< 0.24	≥ 2.5		AC-1s
					> 5.5[d]	AC-1[d]
					2.5–5.5	AC-2z
DS-2	500–1500	400–1400	0.24–0.6	> 3.5		AC-1s
					> 5.5	AC-2
				2.5–3.5		AC-2s
					2.5–5.5	AC-3z
DS-3	1600–3000	1500–3000	0.7–1.2	> 3.5		AC-2s
					> 5.5	AC-3
				2.5–3.5		AC-3s
					2.5–5.5	AC-4
DS-4	3100–6000	3100–6000	1.3–2.4	> 3.5		AC-3s
					> 5.5	AC-4
				2.5–3.5		AC-4s
					2.5–5.5	AC-5
DS-5	> 6000	> 6000	> 2.4	> 3.5		AC-4s
				2.5–3.5	≥ 2.5	AC-5

Notes

a Applies to locations on sites that comprise either undisturbed ground that is in its natural state (ie is not brownfield – Table C2) or clean fill derived from such ground.

b The limits of Design Sulfate Classes based on 2:1 water/soil extracts have been lowered relative to previous Digests (Box C7).

c Applies only to locations where concrete will be exposed to sulfate ions (SO_4) which may result from the oxidation of sulfides (eg pyrite) following ground disturbance (Appendix A1 and Box C8).

d For flowing water that is potentially aggressive to concrete owing to high purity or an aggressive carbon dioxide level greater than 15 mg/l (Section C2.2.3), increase the ACEC Class to AC-2z.

Explanation of suffix symbols to ACEC Class

- Suffix 's' indicates that the water has been classified as static.
- Concrete placed in ACEC Classes that include the suffix 'z' primarily have to resist acid conditions and may be made with any of the cements or combinations listed in Table D2 on page 42.

The sulfide content of the ground must be taken into account if it is concluded that both:

- pyrite is present in significant amounts
- the concrete is to be exposed to disturbed ground (Appendix A1 and Box C8) which might be vulnerable to oxidation.

This procedure should be done in four steps additional to those listed in Section C5.1.1.

Step 6 Determine the characteristic values of the total potential sulfate content for the site location from a consideration of the results of several tests on the pyritic ground. In a data set where five to nine results are available for the location, the mean of the two highest TPS values should be taken as the characteristic value (rounded to 0.1% SO_4). In a data set where 10 or more TPS results are available, the mean of the highest 20% should be taken as the characteristic value.

Step 7 Determine the sulfate class equivalent to the characteristic value of the total potential sulfate content using columns 1 and 6 of Table C2 on the next page.

Step 8 Compare the sulfate class for total potential sulfate with the sulfate classes determined (in Section C5.1.1) for groundwater and water extract tests on soil. The highest of these sulfate classes should then be taken as the Design Sulfate Class for the site location. A limitation can be applied if the sulfate class for the total potential sulfate is initially found to be Sulfate Class 5, but sulfate classes for groundwater and the water extracts tests are Sulfate Class 3 or less. In this case, the Design Sulfate Class for the site location can be limited to DS-4.

The reason for this limitation is that the procedure for sulfate classification based on total potential sulfate is often highly conservative as not all the pyrite in soil will be oxidised and only a part will be taken into solution by groundwater. Some reliance is placed therefore on the findings of field studies of disturbed pyritic clay that has undergone oxidation. These have shown a maximum sulfate class for groundwater in pyritic clay subject to prolonged oxidation to be Sulfate Class 4.

Step 9 Determine the ACEC Class of the ground from the row of Table C1 that correlates first with the Design Sulfate Class, second with the water conditions, and third with the characteristic value of pH.

Table C2 Aggressive Chemical Environment for Concrete (ACEC) classification for brownfield locations [a]

Sulfate and magnesium						Groundwater		ACEC
Design Sulfate Class for location	2:1 water/soil extract [b]		Groundwater		Total potential sulfate [c]	Static water	Mobile water	Class for location
1	2 (SO_4 mg/l)	3 (Mg mg/l)	4 (SO_4 mg/l)	5 (Mg mg/l)	6 (SO_4 %)	7 (pH) [d]	8 (pH) [d]	9
DS-1	< 500		< 400		< 0.24	≥ 2.5		AC-1s
							> 6.5 [d]	AC-1
							5.5–6.5	AC-2z
							4.5–5.5	AC-3z
							2.5–4.5	AC-4z
DS-2	500–1500		400–1400		0.24–0.6	> 5.5		AC-1s
							> 6.5	AC-2
						2.5–5.5		AC-2s
							5.5–6.5	AC-3z
							4.5–5.5	AC-4z
							2.5–5.5	AC-5z
DS-3	1600–3000		1500–3000		0.7–1.2	> 5.5		AC-2s
							> 6.5	AC-3
						2.5–5.5		AC-3s
							5.5–6.5	AC-4
							2.5–5.5	AC-5
DS-4	3100–6000	≤ 1200	3100–6000	≤ 1000	1.3–2.4	> 5.5		AC-3s
							> 6.5	AC-4
						2.5–5.5		AC-4s
							2.5–6.5	AC-5
DS-4m	3100–6000	> 1200 [e]	3100–6000	> 1000 [e]	1.3–2.4	> 5.5		AC-3s
							> 6.5	AC-4m
						2.5–5.5		AC-4ms
							2.5–6.5	AC-5m
DS-5	> 6000	≤ 1200	> 6000	≤ 1000	> 2.4	> 5.5		AC-4s
						2.5–5.5	≥ 2.5	AC-5
DS-5m	> 6000	> 1200 [e]	> 6000	> 1000 [e]	> 2.4	> 5.5		AC-4ms
						2.5–5.5	≥ 2.5	AC-5m

Notes

a Brownfield locations are those sites, or parts of sites, that might contain chemical residues produced by or associated with industrial production (Section C5.1.3).

b The limits of Design Sulfate Classes based on 2:1 water/soil extracts have been lowered from previous Digests (Box C7).

c Applies only to locations where concrete will be exposed to sulfate ions (SO_4), which may result from the oxidation of sulfides such as pyrite, following ground disturbance (Appendix A1 and Box C8).

d An additional account is taken of hydrochloric and nitric acids by adjustment to sulfate content (Section C5.1.3).

e The limit on water-soluble magnesium does not apply to brackish groundwater (chloride content between 12 000 mg/l and 17 000 mg/l). This allows 'm' to be omitted from the relevant ACEC classification. Seawater (chloride content about 18 000 mg/l) and stronger brines are not covered by this table.

Explanation of suffix symbols to ACEC Class

- Suffix 's' indicates that the water has been classified as static.
- Concrete placed in ACEC Classes that include the suffix 'z' have primarily to resist acid conditions and may be made with any of the cements in Table D2 on page 42.
- Suffix 'm' relates to the higher levels of magnesium in Design Sulfate Classes 4 and 5.

Table D1 Selection of the DC Class and the number of APMs for concrete elements where the hydraulic gradient due to groundwater is 5 or less: for general in-situ use of concrete [a,b,c]

ACEC Class (from Tables C1 and C2)	Intended working life At least 50 years[d,e]	At least 100 years
AC-1s, AC-1	DC-1	DC-1
AC-2s, AC-2	DC-2	DC-2
AC-2z	DC-2z	DC-2z
AC-3s	DC-3	DC-3
AC-3z	DC-3z	DC-3z
AC-3	DC-3	DC-3 + one APM of choice
AC-4s	DC-4	DC-4
AC-4z	DC-4z	DC-4z
AC-4	DC-4	DC-4 + one APM of choice
AC-4ms	DC-4m	DC-4m
AC-4m	DC-4m	DC-4m + one APM of choice
AC-5z	DC-4z + APM3[f]	DC-4z + APM3[f]
AC-5	DC-4 + APM3[f]	DC-4 + APM3[f]
AC-5m	DC-4m + APM3[f]	DC-4m + APM3[f]

For specification of DC Class, see Table D2. For choice of additional protective measures, see Table D4.

Notes

a Where the hydraulic gradient across a concrete element is greater than 5, one step in DC Class or one APM over and above the number indicated in this table should be applied except where the original provisions included APM3. Where APM3 is already required, or has been selected, an extra APM is not needed.

b A section thickness of 140 mm or less should be avoided in in-situ construction but, where this is not practical, apply one step higher DC Class or an extra APM except where the original provisions included APM3. Where APM3 is already required, or has been selected, an extra APM is not necessary.

c Where a section thickness greater than 450 mm is used and some surface chemical attack is acceptable, a relaxation of one step in DC Class may be applied. For reinforced concrete, the cover should be sufficiently thick to allow for estimated surface degradation during the intended working life (Section D6.5).

d Foundations of low-rise housing that have an intended working life of at least 100 years may be constructed with concrete selected from the column headed 'At least 50 years' (Section D7).

e Structures with an intended working life of at least 50 years but for which the consequences of failure would be relatively serious, should be classed as having an intended working life of at least 100 years for the selection of the DC Class and APM (Section D7).

f Where APM3 is not practical, see Section D6.1 for guidance.

Explanation of suffix symbols to DC Class

- Concrete placed in ACEC Classes that include the suffix 'z' primarily must resist acid conditions and may be made with any of the cements listed in Table D2.
- Suffix 'm' relates to the higher levels of magnesium in DS Classes 4 and 5.

D5 Composition of concrete to resist chemical attack

D5.1 Background

The main compositional factors that determine the resistance of concrete to aggressive ground are its water/cement ratio and type of cement or combination used. In the previous edition of this Special Digest, the importance of carbonate in the aggregates was stressed in relation to TSA. A source of carbonate is still considered essential for occurrence of TSA, but recent research (Section A3) has shown that sufficient carbonate can come from bicarbonate in groundwater. As a consequence, the limiting values of concrete composition make no distinction between aggregates of different carbonate contents.

Recent research has also shown that resistance to sulfate attack is not a function of cement content. Concretes made with the same materials, and the same w/c ratio but different cement/combination contents, have similar sulfate resistance providing there is sufficient fine material to give a closed microstructure. However, as there is not yet any agreed method for verifying that the concrete has a closed structure, this Special Digest continues to recommend a minimum cement/combination content.

A compressive strength requirement has never formed part of BRE recommendations for sulfate resistance. However, it is recognised that the specification may need to contain a compressive strength class requirement for structural and durability purposes.

Considerable recent research (Section A3) has been focused on determining what is an adequate concrete specification and performance of different cement types. The findings of this research are incorporated into the recommendations given in Table D2. The principal changes as compared with SD1:2003 are:

- the requirements for concrete made with aggregates having a medium or low carbonate content (former aggregate carbonate ranges B or C) have been increased to those given previously for concrete made with aggregates having a high carbonate content (former range A aggregates)
- the excellent performance of concrete incorporating ground granulated blastfurnace slag (ggbs) cements has been recognised and there is some relaxation of the requirements with these cements
- the mixed performance of concrete made with sulfate-resisting Portland cement (SRPC) in sulfate conditions conducive to TSA has led to tightening of requirements

- the performance of concrete incorporating pulverized fuel ash (pfa), and fly ash cements and combinations, is still under investigation and so a conservative approach to their use is taken.

The effectiveness of concretes to resist chemical attack depends to a high degree on their impermeability. Therefore, good compaction is most important. With low w/c ratios, such as those advocated here, it is probable that water reducing admixtures will be needed to achieve effective compaction. This is particularly true of concretes (eg for piling) where mechanical compaction cannot be used.

D5.2 Using Table D2

For a given DC Class, specifications for concrete are shown in Table D2 in terms of maximum free-water/cement or combination ratio and minimum cement or combination content for standard aggregate sizes, and recommended types of cement or combination. The cements and combinations are in new groups, designated A to G, that are defined in Table D3. Table D2 provides a wide range of options for concrete at most DC Class levels so that, in most cases, the concrete producer can use a cement or combination which he normally has in stock.

D5.3 Cement and combination types

D5.3.1 Recommendations in Tables D2 and D3

The cements and combinations specifically recommended by this Special Digest for use in aggressive ground are listed as groups A to G in Tables D2 and D3.

The groups are defined in Table D3 mainly in terms of resistance to sulfate attack. The designations used are based on those of BS EN 197-1 for cement and BS 8500 for combinations. A suffix '+SR' has been added to the designations where a restriction on some element of the composition is necessary for sulfate resistance.

Cements and combinations of the same composition are treated as being directly equivalent and are always grouped together. Moreover, different types (eg CEM II/B-V+SR, a fly ash cement, and CEM III/A+SR, a blastfurnace cement) that show closely similar resistance to sulfate attack are placed in the same Group – in this case, Group D.

While the grouping and nomenclature differ between Table D3 and SD1:2003, in most cases the requirements of cements and combinations with respect to enhanced sulfate resistance remain unchanged.

Table D2 Concrete qualities to resist chemical attack for the general use of in-situ concrete: limiting values for composition

DC Class	Maximum free-water/cement or combination ratio	Minimum cement or combination content (kg/m³) for maximum aggregate size of: ≥ 40 mm	20 mm	14 mm	10 mm	Recommended cement and combination group
DC-1	–	–	–	–	–	A to G inclusive
DC-2	0.55	300	320	340	360	D, E, F
	0.50	320	340	360	380	A, G
	0.45	340	360	380	380	B
	0.40	360	380	380	380	C
DC-2z	0.55	300	320	340	360	A to G inclusive
DC-3	0.50	320	340	360	380	F
	0.45	340	360	380	380	E
	0.40	360	380	380	380	D, G
DC-3z	0.50	320	340	360	380	A to G inclusive
DC-4	0.45	340	360	380	380	F
	0.40	360	380	380	380	E
	0.35	380	380	380	380	D, G
DC-4z	0.45	340	360	380	380	A to G inclusive
DC-4m	0.45	340	360	380	380	F

Grouped cements and combinations

	Cements	Combinations
A	CEM I, CEM II/A-D, CEM II/A-Q, CEM II/A-S, CEM II/B-S, CEM II/A-V, CEM II/B-V, CEM III/A, CEM III/B	CIIA-V, CIIB-V, CII-S, CIIIA, CIIIB, CIIA-D, CIIA-Q
B	CEM II/A-L[a], CEM II/A-LL[a]	CIIA-L[a], CIIA-LL[a]
C	CEM II/A-L[a], CEM II/A-LL[a]	CIIA-L[a], CIIA-LL[a]
D	CEM II/B-V+SR, CEM III/A+SR	CIIB-V+SR, CIIIA+SR
E	CEM IV/B (V), VLH IV/B (V)	CIVB-V
F	CEM III/B+SR	CIIIB+SR
G	SRPC	–

For cement and combination types, compositional restrictions and relevant Standards, see Table D3.

Note

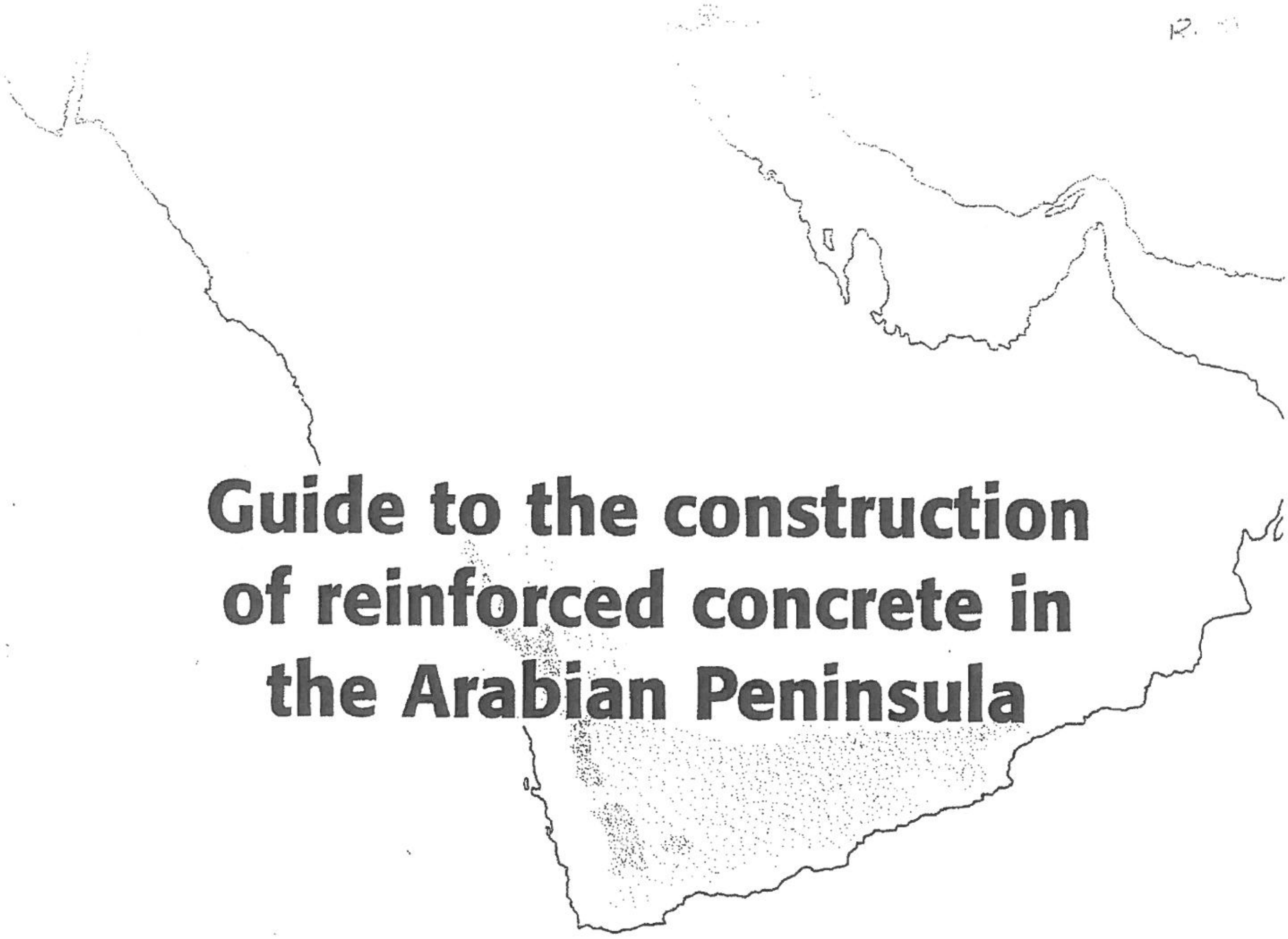
a The classification is B if the cement/combination strength class is 42,5 or higher and C if it is 32,5.

Table D3 Cements and combinations for use in Table D2

Type	Designation	Standard	Grouping with respect to sulfate resistance
Portland cement	CEM I	BS EN 197-1	A
Portland-silica fume cement	CEM II/A-D	BS EN 197-1	A
Portland-limestone cement	CEM II/A-L	BS EN 197-1	B[a] or C[a]
	CEM II/A-LL	BS EN 197-1	B[a] or C[a]
Portland-pozzolana cement	CEM II/A-Q[b]	BS EN 197-1	A
Portland-slag cements	CEM II/A-S	BS EN 197-1	A
	CEM II/B-S	BS EN 197-1	A
Portland-fly ash cements–	CEM II/A-V	BS EN 197-1	A
	CEM II/B-V[c]	BS EN 197-1	A
	CEM II/B-V+SR[d]	BS EN 197-1	D
Blastfurnace cements[e]	CEM III/A	BS EN 197-1	A
		BS EN 197-4	A
	CEM III/A+SR[f]	BS EN 197-1	D
		BS EN 197-4	D
	CEM III/B	BS EN 197-1	A
		BS EN 197-4	A
	CEM III/B+SR[f]	BS EN 197-1	F
		BS EN 197-4	F
Pozzolanic cement[g,h]	CEM IV/B (V)	BS EN 197-1	E
Very low heat pozzolanic cement	VLH IV/B (V)	BS EN 14216	E
Sulfate-resisting Portland cement	SRPC	BS 4027	G
Combinations conforming to BS 8500-2, Annex A, manufactured in the concrete mixer from Portland cement and fly ash, pfa, ggbs or limestone fines:			
CEM I cement conforming to BS EN 197-1 with a mass fraction of 6 to 20 % of combination of fly ash conforming to BS EN 450 or pfa conforming to BS 3892-1	CIIA-V	BS 8500-2, Annex A	A
CEM I cement conforming to BS EN 197-1 with a mass fraction of 21 to 35 % of combination of fly ash conforming to BS EN 450 or pfa conforming to BS 3892-1	CIIB-V[c]	BS 8500-2, Annex A	A
	CIIB-V+SR[d]	BS 8500-2, Annex A	D
CEM I cement conforming to BS EN 197-1 with a mass fraction of 36 to 55 % of combination fly ash conforming to BS EN 450 or pfa conforming to BS 3892-1	CIVB-V	BS 8500-2, Annex A	E
CEM I cement conforming to BS EN 197-1 with a mass fraction of 6 to 35 % of combination of ggbs conforming to BS 6699	CII-S	BS 8500-2, Annex A	A
CEM I cement conforming to BS EN 197-1 with a mass fraction of 36 to 65 % of combination of ggbs conforming to BS 6699	CIIIA	BS 8500-2, Annex A	A
	CIIIA+SR[f]	BS 8500-2, Annex A	D
CEM I cement conforming to BS EN 197-1 with a mass fraction of 66 to 80 % of combination of ggbs conforming to BS 6699[e]	CIIIB	BS 8500-2, Annex A	A
	CIIIB+SR[f]	BS 8500-2, Annex A	F
CEM I cement conforming to BS EN 197-1 with a mass fraction of 6 to 20 % of combination of limestone fines conforming to BS 7979	CIIA-L	BS 8500-2, Annex A	B[a] or C[a]
	CIIA-LL	BS 8500-2, Annex A	B[a] or C[a]
CEM I cement conforming to BS EN 197-1 with a mass fraction of 6 to 10 % of combination of silica fume conforming to BS EN 13263[i]	CIIA-D	See Note j	A
CEM I cement conforming to BS EN 197-1 with a mass fraction of 6 to 20 % of combination of metakaolin conforming to an appropriate Agrément certificate	CIIA-Q	See Note k	A

Notes

a The classification is B if the cement or combination strength is class 42,5 or higher and C if it is class 32,5.
b Metakaolin only.
c Where the fly ash or pfa content is a mass fraction of 21 to 24%.
d The addition of the abbreviation '+SR' denotes an additional requirement for sulfate resistance that the fly ash content should be a mass fraction of not less than 25% of the cement or combination. Where it is less than 25%, the grouping with respect to sulfate resistance is 'A' (Note c).
e Cements or combinations with higher levels of slag than permitted in this table may be used for certain specialist applications, but no guidance is provided in this Special Digest or BS 8500.
f The addition of the abbreviation '+SR' denotes an additional requirement for sulfate resistance, that where the alumina content of the slag exceeds 14%, the tricalcium aluminate content of the Portland cement fraction should not exceed 10%. Where this is not the case, the grouping with respect to sulfate resistance is 'A'.
g CEM IV/A cement with siliceous fly ash should be classified as CEM II-V cement.
h (V) indicates siliceous fly ash only.
i Until BS EN 13263 is published, the silica fume should conform to an appropriate British Board of Agrément certificate.
j These combinations are not currently covered by BS 8500-2, Annex A. However, silica fume can be used in accordance with Clause 5.2.5 of BS EN 206-1.
k These combinations are not currently covered by B S 8500-2, Annex A. However, metakaolin conforming to Clause 4.4 of BS 8500-2 may be used in accordance with Clause 5.2.5 of BS EN 206-1. If the *k*-value concept is used, a *k*-value with respect to sulfate resistance of 1.0 should be used.



Guide to the construction of reinforced concrete in the Arabian Peninsula

CIRIA

Editor: Mike Walker

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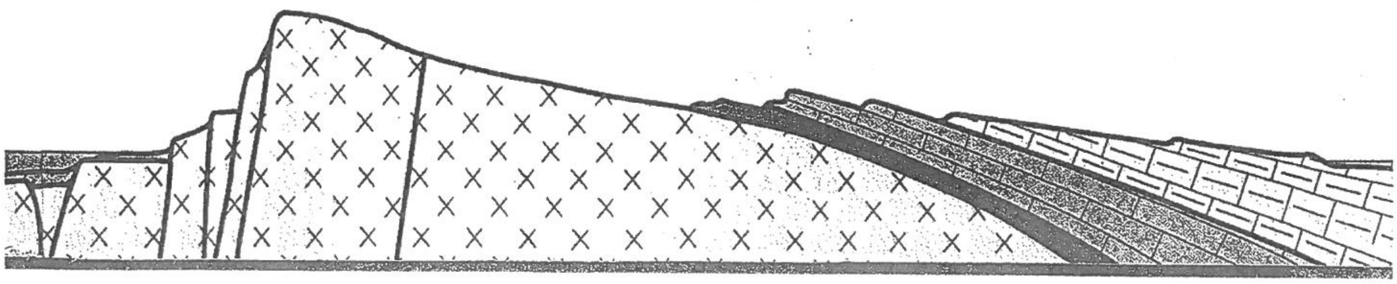

Table 5.1: *Classification of exposure conditions in the Arabian Peninsula.*

Exposure condition	Locations
a	Superstructures inland with no risk of windborne salts
b	Superstructures in areas of salt flats, inland or near the coast, exposed to windborne salts
c	Parts of structures in contact with the soil, well above capillary rise zone, with no risk of water introduced at the surface by irrigation, faulty drainage systems, washing down etc.
d	Parts of structures in contact with the soil, within the capillary rise zone, below groundwater level, or where water may be introduced at the surface by irrigation, discharge of wastes, washing down etc. These situations all lead to a potential for the concentration of aggressive salts by evaporation. (i) Significant sulfate contamination only (ii) Significant chloride contamination only (iii) Significant contamination with both sulfates and chlorides
e	Marine structures (splash zone)
f	Water-retaining structures (including sewage treatment plants)

Cements

The properties of cement should be independently tested to ensure that it is suitable for its purpose, and that it complies with ASTM or BSI standards or their equivalent. (See Appendix 2 for notes about the introduction of a new European nomenclature for cements and their British Standard equivalents.

- Selection of cement type is covered in detail in Chapter 9. This includes information on combinations of cementitious materials.
- Tables 9.7 and 9.9 summarise specification requirements for cement and Table 9.1 gives characteristics of different types of Portland and blended cements.

The type of cement to be used depends on the function and exposure of the concrete.

- Tables 5.1 and 5.2 give information on typical exposure conditions and recommended mix and cover criteria.

Further information on the use of materials such as pulverised-fuel ash, ground granulated blastfurnace slag and silica fume, sometimes referred to as mineral additions, is given in the appropriate chapters.

Water

It is important that water for mixing or curing reinforced concrete is substantially free from contamination, particularly that which would raise the chloride or sulfate content of the concrete.

- Table 11.1 recommends limits for principal contaminants in mixing-quality water.

Admixtures

Plasticisers are usually needed in all mixes to achieve sufficient workability with high quality concrete in hot climates. Air-entraining agents are recommended for reducing settlement and bleeding where aggregate gradings are deficient in some sizes.

- Table 10.1 summarises performance requirements for different types of admixtures.

Concrete mix design

Concrete must be workable enough when it leaves the mixer to allow for loss of workability in hot weather and to still be thoroughly compacted between and around the reinforcement. Concrete mixes should contain enough cement (in combination with superplasticisers if appropriate) to give this workability with a low water/cement ratio, so that the concrete has a closed pore structure and a low permeability to aggressive salts etc.

- Table 5.2 summarises typical concrete mix criteria specified in the Gulf region.

Table 5.2: *Typical concrete mix criteria and cover requirements for exposure conditions in the Arabian Peninsula, from Table 5.1.*

Exposure conditions	Cementitious material(s)	Minimum cementitious content for 20 mm aggregates (kg/m^3)	Maximum free-water/cement ratio **	Additional requirements	Minimum cover to reinforcement (mm)
a	Portland cement or additions	300–320	0.52	None	30
b		320	0.50	None	40
c*		320–350	0.45	None	40–50
d(i), (ii) or (iii)		320–400	0.42	Tanking	40–50
e and f	Portland cement blends with additions	370–400	0.40	None	100–150

* When concrete is cast directly in contact with soil the minimum cover should be increased to 75 mm.

** On well supervised projects free-water/cement ratios down to 0.35 have been successfully achieved using the latest generation of superplasticisers.

The coating should be aesthetically pleasing and durable, and resist carbon dioxide ingress. Generally, pigmented coatings are more durable than non-pigmented. The service life of a structure may be expected to exceed that of the coating, so provision should be made during project design for future maintenance and/or re-application of the coating. Advice should be sought from coating manufacturers on suitable maintenance periods.

As the surface of concrete may not be very smooth, applying effective coatings can be difficult. Thick coatings may be needed to ensure that all the discontinuities and protrusions in the surface are adequately covered. With smooth surfaces, thinner coatings will suffice. The use of coatings to resist chloride ingress is included in Section 6.5.

6.2.3 Preventing chloride ingress into concrete

The penetration of chloride into concrete is difficult to avoid in chloride-laden atmospheres. In the region, the practice of using Portland cement concrete with a water/cement ratio below 0.45 with reinforcing steel at least 75 mm from the surface, does not always prevent chloride-induced corrosion of embedded steel. To prevent corrosion, the level of chloride at the steel needs to be minimised, i.e. the penetration rate reduced, as summarised in Table 6.1.

Chloride penetration

As outlined in Chapter 2, resistance to chloride penetration is influenced by the cement chemistry and concrete quality. In general, Portland cement with a high C_3A content is more resistant to chloride penetration than Portland cement with a low C_3A content. This is partly due to the influence of alumina in the binding of some chloride during mixing and placing. Differences in Portland cement composition have been reported to result in differences in chloride diffusion coefficient of the order of 10 times (Frey *et al.*, 1994).

Concretes containing pulverised-fuel ash (pfa), ground granulated blastfurnace slag (ggbs), and silica fume (sf) are highly resistant to penetration by chlorides, due to their increased binding capacity and refined pore structure. Service lives two to three times longer are estimated for concretes made with pfa or ggbs of equal water/cementitious material ratio. This difference is greater when comparisons are made on the basis of strength grade, because blended

Table 6.1: *Main methods of reducing the penetration of chlorides.*

Approach	Method
Concrete mix design	Selection of cement type Water/cement ratio Use of additions: Pulverised-fuel ash Ground granulated blastfurnace slag Silica fume
Other measures	Controlled permeability formwork Coatings Hydrophobic treatment of the concrete

cement mixes need a lower water/cement ratio to achi strength equivalence at 28 days. However, care must be ta when evaluating pfa and ggbs concretes, as their resistanc chloride develops more slowly than for Portland cement silica fume concrete. A short-term accelerated test may fai demonstrate any appreciable benefits.

Normal concretes containing silica fume will reduce chlori penetration. When used in conjunction with a very low wat cement ratio, the potential increase in resistance to chlori penetration is considerable.

Surface chloride levels

When predicting the time to corrosion, chloride content at tl surface is taken into account (Bamforth and Price, 1993). should be noted that surface chloride levels tend to be high in concretes with a higher binding capacity and a highe resistance to chloride penetration. For example, ggbs con cretes tend to have surface chloride contents about 15% higher than Portland cement concretes, as chlorides are slowe to penetrate into the concrete thus causing the surface chloride level to build up. This is offset by the lower diffusion coef- ficient in the body of concrete containing such additions as pfa, ggbs, and sf.

Threshold levels

The chloride threshold level for the initiation of corrosion depends on many factors, including the cement type. In Portland cement concretes, the C_3A content has a significant effect on chloride threshold levels mainly because of the high binding capacity of cements higher in C_3A. A Portland cement with a moderate C_3A content (of 8%) may have a threshold level double that of sulfate-resisting Portland cement with a low C_3A content (of 3.5%). Alkalinity is also important, with higher values of chloride being acceptable at high pH. Mineral additions tend to lower pH, as they react with the alkalis from Portland cement. At lower pH, the binding capacity of the cement is increased, but the reduction in free chlorides is less than the reduction in hydroxyl ions. Because the chloride/hydroxyl ion ratio increases as the pH reduces, blended cements tend to have lower threshold levels than Portland cements. Temperature also has a dominant effect on the threshold level with lower values at higher temperatures. Compared with values at 20°C, the threshold level may reduce by 50% at 40°C and by more than 80% at 70°C (Benjamin and Sykes, 1989; Hussain *et al.*, 1995).

Selection of cements for chloride resistance

There have been cases where very low threshold levels (0.15% by weight of cement) have been observed in the Arabian Peninsula (Rasheeduzzafar *et al.*, 1985), illustrating once again that precise values of threshold are difficult to predict. It is therefore advisable to use a cementing system that prevents any chlorides from penetrating to the depth of the steel within the proposed life of the structure.

Normal grade structural concretes containing pfa, ggbs or sf mixes can all achieve very high chloride resistance compared with Portland cement concretes.

9.3.1 Portland cement

Portland cements consist of oxides of four main elements (calcium, silicon, aluminium and iron) and several minor constituents. These are generally abbreviated, using 'cement chemist's shorthand': A for Al_2O_3, aluminium oxide (alumina); C for CaO, calcium oxide (lime); F for Fe_2O_3, ferric oxide; S for SiO_2, silicon oxide (silica). Likewise, H is used for H_2O, water.

The major constituents are combined to form four mineral compounds simplified using the 'shorthand' as:

- C_3A tricalcium aluminate (alite)
- C_2S dicalcium silicate (belite)
- C_3S tricalcium silicate
- C_4AF tetracalcium alumino ferrite (celite).

A typical oxide composition for Portland cement is shown in Table 9.2. The properties of these compounds are also included in Table 9.2 and repeated in Table 9.3 to demonstrate how changes in their proportions can influence the properties of cement (Soroka, 1993).

In practice, the usual reason for using cement other than Portland cement is either to reduce chemical attack on the concrete, to improve durability, or to change the rate of gain

Table 9.1: *Characteristics of different types of Portland and blended cements.*

Type of cement	Relevant specifications	Resistance to chemical sulfate attack	Protection of reinforcement against chloride-induced corrosion	Rate of gain of strength and rate of heat evolution
Portland cement or ASTM Type I	BS 12 ASTM C150	Not resistant: aggregates should be virtually free from sulfates and concrete should be isolated from groundwaters	The protection given by this cement is taken as 'normal' for comparative purposes	Taken as 'normal' for comparative purposes
ASTM Type II (no BS equivalent)	ASTM C150	Partially resistant	If this cement contains 5% or more of C_3A it may give near normal protection	Lower than Portland cement
Rapid-hardening Portland cement or ASTM Type III	BS 12 ASTM C150	Same as Portland cement	Same as Portland cement	Hydration usually too rapid for use in hot climates
Low-heat Portland cement or ASTM Type IV	BS 1370 ASTM C150	May be more resistant than Portland cement if C_3A content is lower, but the specification does not guarantee this.	Less than normal if C_3A content is lower than that of Portland cement	Lower rate of heat evolution, strength gain and lower final strength than Portland cement
Sulfate-resisting Portland cement or ASTM Type V	BS 4027 ASTM C150	Moderately resistant	This cement contains little or no C_3A and gives significantly less protection against chloride-induced corrosion than Portland cement or ASTM Types I or II	Lower rate of heat evolution and strength gain than Portland cement or Type I, but usually not as low as low-heat Portland cement or ASTM Type IV
High slag blastfurnace cement	BS 4246	Properties vary but sulfate resistance will be equivalent to Type V or srpc if 70% or more ggbs is used	Low permeability to chloride and high binding capacity	Gains strength slower but continues to gain strength at later ages. Low heat evolution equivalent or lower than low-heat Portland cement Type IV depending on ggbs content
Portland blastfurnace cement and low-heat Portland blastfurnace cement	BS 146	Properties vary, but sulfate resistance could be equal to or better than ASTM Type II	Good impermeability to chloride penetration and binding capacity depending on ggbs content	Gains strength and evolves heat more slowly than Portland cement but not as slowly as low-heat Portland cement or ASTM Type IV, unless made to a low-heat specification
Other cements containing ggbs or pozzolana incorporated in the cement during manufacture	The properties of these cements depend on the materials they are made from and on the proportions of ggbs or pozzolana incorporated in them.			

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MINISTRY OF INDUSTRY
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Conformity Assessment Body's ID: NAL 225

الرمز التعريفي لجهة تقييم المطابقة: NAL 225

CONSULT SOIL TESTING LABORATORY
SHOP-7, BEHIND ZEIN ITALIA MARBLES, QUSAIDAT, Ras Al-Khaimah, United Arab Emirates

مختبر كونسلت لفحص التربة
متجر رقم-7,خلف زين ايطاليا للرخام,القصيدات، رأس الخيمة، الإمارات العربية المتحدة

Accredited according to the International Standard ISO/IEC 17025_1 to undertake tests as specified in the attached Accreditation Scope.

حاصل على الإعتماد لطرق الفحص الواردة في وثيقة المجال المرفقة وفقاً للمواصفة الدولية ISO/IEC 17025_1

Accredited on: 28/11/2024

تاريخ منح الاعتماد: 28/11/2024

Expires on: 27/11/2027

تاريخ الانتهاء: 27/11/2027

1 Accreditation in accordance with the International Standard ISO/IEC 17025:2017 "General requirements for the competence of testing and calibration laboratories" and the relevant ENAS and ILAC requirements.

1 وفقاً لمتطلبات المواصفة الدولية ISO/IEC 17025:2017 "المتطلبات العامة لكفاءة مختبرات الفحص والمعايرة" والمتطلبات ذات العلاقة الخاصة بنظام الإعتماد الوطني الإماراتيENAS والمنظمة الدولية لاعتماد المختبرات ILAC.

This certificate is invalid without the attached scope of accreditation, which subjected to annual surveillances as per ENAS procedure. Certificate can be updated or re-issued until the expiry date defined above. The validity of the certificate is subjected to continuous compliance with the requirements of the accreditation system. The lab is responsible for the results of its testing.

مجال الاعتماد جزء أساسي من هذه الشهادة حيث تخضع مجالات الاعتماد المذكورة في الوثيقة المرفقة لعمليات متابعة لاحقة من قبل نظام الإعتماد الوطني الإماراتي ENAS، وتعتبر هذه الشهادة صالحة وقابلة للتحديث واعادة الاصدار حتى تاريخ الانتهاء المدون اعلاه شريطة استمرار المختبر المذكور اعلاه في تطبيق متطلبات نظام الاعتماد سالفة الذكر. يتحمل المختبر مسؤولية نتائج الفحص الصادرة عنه.

This is an electronic certificate and does not require stamp and signature. Certificate will be invalid in case of any modification.

هذه الشهادة صدرت إلكترونياً ولا تحتاج لختم أو توقيع، أي كشط أو تغيير في هذه الشهادة يلغيها.

Initial Accreditation Date: 28/11/2024

تاريخ منح الإعتماد لأول مرة: 2024/11/28

Ras Al Khaimah Municipality – Urban Development Sector

Permitting Department

Third-Party Laboratory Registration Certificate

In compliance with Ras Al Khaimah Controls on Third-party Laboratories serving the construction works in the Emirate of Ras Al Khaimah, Ras Al Khaimah Municipality – Urban Development Sector - Permitting Department hereby issue this registration certificate to:

Client Name:	**CONSULT SOIL TESTING LABORATORY**
Address:	401090044, P. O. Box: 33948, Al Tarafa Street, RAK, UAE

Reference Accreditation Certificate to ISO 17025/Special entity approval

Accreditation Body/ Specialized Entity	Ministry of Industry and Advanced Technology (MOIAT- ENAS)
Accreditation Scope:	SOIL & CONSTRUCTION MATERIALS*
Accreditation Certificate #:	NAL 225
Accreditation Expiry Date:	27/11/2027

* Only hardened concrete compressive strength testing for construction materials.

Registration Number:	TPLR25040003
First Issue Date:	28/02/2025
Current Issue Date:	28/02/2025
Expiry Date:	27/02/2026

Ras Al Khaimah Municipality – Urban Development Sector - Permitting Department

يعتبر مجال شهادة التسجيل المرفق جزءاً لا يتجزأ منها ويجب قراءتهما معاً

The scope of Registration Certificate is an integral part of the certificate and shall be read together